

Quantifying the Limits:

A Systematic Evaluation of Post-Processing Popularity Bias Mitigation Under Varying Base-Model Bias

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Abstract

Recommender systems frequently exhibit popularity bias, where a small number of highly popular items receive a disproportionate share of exposure. Post-processing reranking methods are widely used to mitigate this bias because they are model-agnostic and require no retraining of the base recommender. However, existing research evaluates these methods under fixed bias conditions, leaving open the question of how their effectiveness changes as the underlying base-model bias varies.

This thesis addresses that gap through a controlled two-stage experimental framework. Popularity bias is systematically induced at five levels using power-law interaction resampling ($\alpha \in \{1.0, 1.2, 1.5, 1.75, 2.0\}$) and applied to two benchmark datasets (MovieLens-1M and Amazon Books). Three post-processing methods, Binary xQuAD, Smooth xQuAD, and FA*IR, are evaluated on top of three base model architectures (BPR, NeuMF, LightGCN), with parameters selected using a Nash bargaining criterion.

The central finding is the identification of a *structural ceiling*: a hard upper bound on achievable long-tail exposure that declines monotonically with base-model bias, independently of the reranking method applied. This ceiling arises from the progressive depletion of long-tail items in the candidate pool as the base model concentrates its representations on head items. All three methods improve long-tail coverage at every bias level, but the maximum achievable gain shrinks as bias increases, and the ceiling is more severe for architectures with front-depleted candidate pools. Nash-selected parameters remain stable across all bias levels and transfer across architectures without retuning. The results demonstrate that post-processing alone cannot compensate for severe base-model bias, and that effective long-tail promotion requires coordinated intervention at the retrieval stage as well as the reranking stage.

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Chapter 1

Introduction

1.1 Motivation and Context

Recommender systems have become a core part of how people interact with digital platforms. Whether it is finding a new show to watch, discovering music, or browsing products online, these systems shape what users see and, by extension, what they engage with. For many platforms, recommendations are not just a convenience feature; rather, they are responsible for a significant share of user interactions and revenue, making them central to both user experience and business value. Although these systems are essential for navigating digital content, they are often prone to different biases that can undermine the very discovery they are meant to facilitate [Chen et al., 2023].

Many recommendation models exhibit popularity bias, a tendency to favor globally popular items disproportionately over less popular or niche content. Instead of broadening access to less frequently consumed items (long tail 1.1), recommender systems often focus recommendations on small items that are frequently consumed (head 1.1) [Abdollahpouri et al., 2019b].

The implications of popularity bias extend across multiple stakeholder groups. For users, it translates into a reduced capacity for meaningful discovery and a degradation of perceived personalization quality. For content creators, particularly those whose work lacks an established audience, it represents a structural disadvantage that is largely independent of content quality. For platform operators, it raises questions about ecosystem sustainability and excessive dependence on a narrow pool of content [Carnovalini et al., 2025].

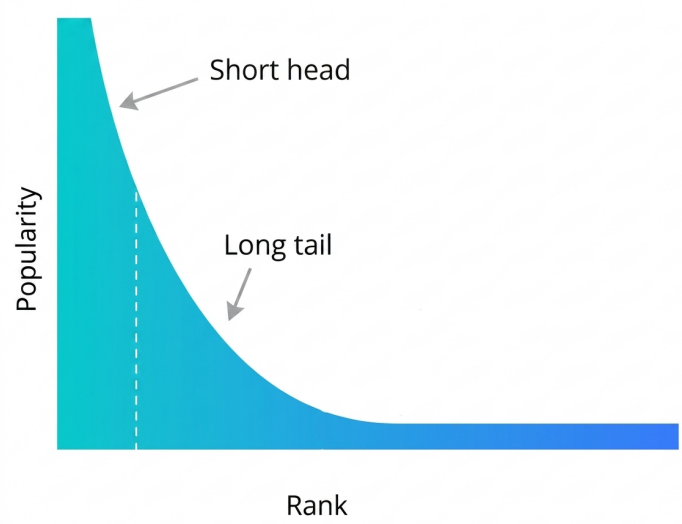


Figure 1.1: The head and long-tail distribution of item popularity from the recommender list (adapted from [Abdollahpouri et al., 2019a]).

1.2 Problem Statement

Popularity bias in recommender systems arises from multiple interconnected sources. At its core, the phenomenon reflects genuine human tendencies: popular items often become popular for valid reasons, and user behavior naturally gravitates toward familiar choices [Klimashevskaia et al., 2023]. However, algorithmic systems can amplify these tendencies far beyond their natural occurrence, creating distortions that serve neither users nor providers optimally.

Bias amplification by the recommender model is particularly evident in collaborative filtering approaches, where [Zhu et al., 2021] identify "popularity-opportunity bias" as the tendency for systems to recommend popular items over equally relevant but less popular alternatives when users would be interested in both. Their theoretical analysis demonstrates that matrix factorization models inherently amplifies this bias, suggesting that the problem is not merely an implementation artifact but a fundamental characteristic of widely-used recommendation approaches.

Several approaches have been developed to reduce popularity bias in recommendation systems [Wang et al., 2022]. Current mitigation strategies broadly fall into three categories:

- **Pre-processing:** Modifying training data.
- **In-processing:** Incorporating fairness constraints during model train-

ing.

- **Post-processing:** Adjusting final recommendations.

Each approach offers distinct advantages and trade-offs, but post-processing methods have gained particular attention due to their model-agnostic nature and practical deployability. Methods such as personalized re-ranking [Abdollahpouri et al., 2017], calibration-based approaches [Souza et al., 2024], and specific re-ranking algorithms [Machado et al., 2024] demonstrate that post-processing can effectively reduce bias without requiring fundamental changes to existing infrastructure.

However, a critical gap exists in our understanding of the effectiveness of these methods under varying conditions. While research demonstrates that post-processing can mitigate bias, it provides limited insight into how this effectiveness changes as the underlying bias in base recommendation models (e.g., BPR vs. Neural approaches) varies.

1.3 Research Gap and Limitations

Despite the growing body of research, significant limitations constrain our understanding of when and how post-processing methods succeed or fail. [Klimashevskaia et al., 2023] note that current research is "almost entirely based on computational experiments" with limited validation of practical effects. More critically, [Carnovalini et al., 2025] observe that "some current approaches fail to face the problem in its entirety," suggesting gaps in evaluation and deployment strategies.

The most significant gap lies in the **systematic evaluation of post-processing effectiveness under varying levels of base-model bias**. Existing studies typically evaluate mitigation methods against fixed baseline systems. However, real-world systems exhibit varying degrees of bias depending on data sparsity, algorithm choice, and training procedures.

This limitation has profound practical implications. Practitioners lack guidance on how a method that performs well on a moderately biased system might over-correct or prove inadequate when applied to highly biased baselines. Furthermore, the fairness-utility trade-offs remain poorly characterized across different bias conditions. While [Wang et al., 2022] discuss trade-offs between fairness and accuracy, these findings are rarely reported across systematic variations in base-model bias.

1.4 Research Questions and Contributions

This thesis addresses the identified gap through a systematic evaluation of post-processing popularity bias mitigation methods under varying levels of base-model bias. The research is guided by three interconnected questions that collectively aim to quantify the limits and opportunities of post-processing approaches.

RQ1: Measure Bias Impact - How do different levels of popularity bias in the base recommender affect the performance of post-processing mitigation methods? This question establishes the fundamental relationship between base-model bias and mitigation effectiveness, providing the empirical foundation for understanding method limitations.

RQ2: Map Trade-offs - How do fairness–utility trade-offs change as base bias increases? Building on RQ1, this question characterizes how the fundamental tension between recommendation accuracy and bias mitigation evolves across different bias conditions, informing optimal deployment strategies.

RQ3: Compare Methods and Architectures — How robust are different post-processing strategies under rising bias across datasets and base model architectures? This question evaluates the relative strengths and weaknesses of existing approaches, examining both how reranking methods differ in their resilience and how the choice of base model architecture shapes the ceiling within which all methods must operate.

Based on these questions, the study is guided by the following hypotheses:

- **H1:** The effectiveness of fairness-aware reranking methods decreases as popularity bias α increases.
- **H2:** Different reranking strategies exhibit varying robustness to increasing bias levels.
- **H3:** The optimal fairness–utility trade-off parameter may depend on the underlying bias distribution.

The primary contribution of this work is the identification and characterization of a *structural ceiling* on post-processing popularity bias mitigation: a hard, data-driven upper bound on achievable long-tail exposure that declines monotonically with the level of base-model bias, independently of the reranking method applied. This ceiling arises from the progressive depletion of long-tail items in the candidate pool as the base model concentrates its representations on head items, and it establishes that the limiting factor for fairness-aware reranking lies upstream of the reranker itself.

Secondary contributions include a controlled two-stage experimental framework for inducing and measuring base-model bias at five levels across two datasets and three architectures; a Nash bargaining criterion for principled, reproducible parameter selection that remains stable across bias conditions; a characterization of how base model architecture independently determines the severity of the structural ceiling; and practical guidelines for practitioners on the limits of post-processing and the need for coordinated bias mitigation at the training stage.

1.5 Thesis Structure and Outline

This thesis is organized into seven chapters. Chapter 2 reviews related work on popularity bias and post-processing mitigation methods. Chapter 3 presents the methodology, including the bias induction procedure, the candidate-pool framework, and the Nash-based parameter selection criterion. Chapter 4 reports the experimental results across all five bias levels, two datasets, and three architectures. Chapter 5 discusses the results in relation to the three research questions and revisits the hypotheses. Chapter 6 concludes with a summary of contributions, limitations, and directions for future work.

Chapter 2

Related Work

2.1 Popularity Bias

Although recommender systems are intended to help users in dealing with information overload, they often show a consistent bias toward popular items. This phenomenon is referred to as popularity bias. Popularity Bias arises particularly when popular items are recommended more often than their actual popularity would warrant [Abdollahpouri and Mansoury, 2020]. Essentially, the algorithmic amplification of existing popularity patterns results in a self-reinforcing cycle in which items that are already popular receive disproportionate visibility.

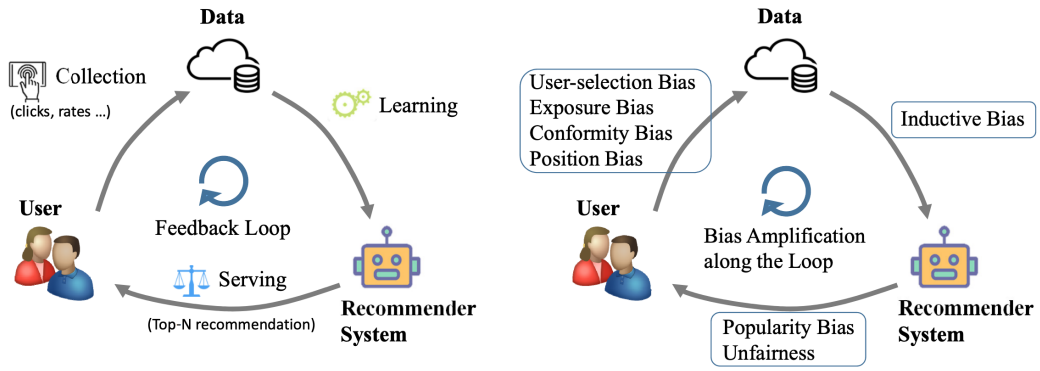


Figure 2.1: Sources and amplification of bias (adapted from [Chen et al., 2023]).

Sources and Amplification Mechanisms

Popularity bias is not an isolated artifact but an emergent property of the **recommendation feedback loop** . In general, biases propagate through three stages: User → Data (collection), Data → Model (learning), and Model → User (serving) [Chen et al., 2023]. At each stage, bias can enter and subsequently be amplified. Understanding this causal structure is essential because it explains why popularity bias persists in recommender systems 2.1.

Data-level biases reflect the natural, pre-existing skew in user interaction patterns. Usually, popularity is directly connected with interaction count in recommender systems. If an item has more user interactions, then it is seen as a popular item. As [Klimashevskaia et al., 2024] note, items may be more popular due to genuine quality differences, pricing advantages, or promotional activity, providing the initial seed for algorithmic amplification. On the other hand, not all observed popularity reflects genuine quality. Interaction counts of items can be influenced by presentation order of items by the service provider, marketing spending on items, and historical factors. The observational nature of user data means that users are only exposed to a subset of items, so unobserved interactions do not always represent negative preference [Chen et al., 2023].

Model-level biases arise from the inductive biases of learning algorithms. Recommender models that uses collaborative filtering as their main approaches have proven to generate final recommended list that favor popular items. [Zhu et al., 2021] identify "popularity-opportunity bias"—which is the tendency for systems to recommend popular items over equally relevant but less popular alternatives. [Zhu et al., 2021] demonstrate that for example, matrix factorization models inherently produce this "popularity-opportunity bias", suggesting it is not merely an implementation artifact but a fundamental characteristic of widely-used approaches. [Chen et al., 2023] further explain that optimization objectives of many recommender models implicitly reward predicting popular items accurately, even when those predictions are trivial.

The feedback loop effect intensifies biases over time. As [Chen et al., 2023] observe, the exposure mechanism of a recommender system influences user behaviors, which are then fed back as training data for a recommender model. In the end this creates a "Matthew effect" problem where popular items become progressively more dominant. Simulation studies confirm that the feedback loop amplifies popularity bias, making popular items even more

popular while non-popular items become even less visible [Mansoury et al., 2020b].

Relationships with Other Quality Aspects

Popularity bias intersects with multiple beyond-accuracy objectives, though these relationships are often oversimplified [Klimashevskaia et al., 2024]. The three widely acknowledged beyond-accuracy objectives include Novelty, Diversity, and Fairness.

Novelty: It can be defined as the degree to which a recommender model is able to introduce new items to users. Novelty can be seen as being inversely related to popularity. It is common to formulate technical realizations of novelty measures as inversely connected to popularity metrics [Castells et al., 2021]. It is important to understand that just because Novelty is inversely connected to popularity, then it can be seen as a technique for debiasing popularity bias in recommender model. This is because, for example, an item can be novel to a specific user while still being globally popular, and conversely, a long-tail item may already be known to the user. The relationship is probabilistic, not deterministic.

Diversity: It can be defined as the extent to which the recommended items differ from each other. Diversity tends to be affected by popularity bias but not uniformly. Further more, diversity can be explored as *Aggregate diversity* or *catalog coverage*, which basically accounts for the fraction of items that appear in recommendation lists. This beyond-accuracy objective is the most directly impacted by popularity bias [Adomavicius and Kwon, 2011].

Fairness represents the most ethically charged connection. [Klimashevskaia et al., 2024] argue that simply increasing the exposure of arbitrary items does not necessarily serve a fairness goal. This is because generally some items may be unpopular due to limited general appeal or quality. Fairness requires a specific justification for why certain groups deserve particular treatment and this vary based on the fairness objective.

Stakeholder Impacts

The negative effect of popularity bias can be seen in multiple stakeholder groups.

Users: For users, particularly those with niche preferences, popularity bias translates into poor recommendation quality. This is because these users never get to see niche items that they would prefer to see [Ekstrand et al., 2018].

Content creators and item providers: Popularity bias creates structural disadvantages for this group. For example, new service providers struggle to break through in a new market due to saturation of popular providers despite providing equal or even better services. A specific example of this problem is in music streaming services. [Ferraro et al., 2021] explores the gender imbalance in music recommendation and shows how a male artist generally dominates the list of recommended artists compared to a female artist. This is further confirmed by [Dinnissen and Bauer, 2023] as they explore how artists perceive fairness on streaming platforms.

Platform operators: [Lee and Hosanagar, 2019] found in a large-scale field experiment that while a recommender increased sales of niche items, the increase for popular items was even more pronounced and the diversity of aggregate sales actually decreased.

The Double-Edged Nature of Popularity

Although popularity bias has a negative effect on different stakeholders, it is not universally harmful. In their work, [Chen et al., 2023] explicitly acknowledge that popularity bias has been validated as a double-edged sword. For example, in cold-start scenarios, recommending popular items represents a sensible default strategy [Rashid et al., 2002], and industrial systems such as Netflix deliberately incorporate popularity signals [Gomez-Uribe and Hunt, 2015]. [Zhao et al., 2022] argue that the goal should be calibrating popularity bias, not eliminating the popularity signals entirely.

2.2 Mitigation Taxonomy

Over the years, there has been increasing acknowledgment of how popularity bias adversely affects different stakeholders. In response to this, there has also been an increase in proposals of mitigation techniques. [Klimashevskaia et al., 2024] Classifies these approaches using a three-stage framework which is based on when the mitigation intervention takes place:

- **Pre-processing approaches**: Interventions applied to the training data before model learning.
- **In-processing approaches**: Modifications integrated into the model's training objective or architecture.
- **Post-processing approaches**: Adjustments made to ranked output lists after initial recommendation generation.

This taxonomy has gained widespread acceptance because it captures fundamental design decisions with practical implications for deployment.

Distribution of Approaches in the Literature

The literature on popularity bias mitigation techniques shows that most of these approaches belong to in-processing approaches, followed by post-processing methods, and pre-processing having the most least techniques [Klimashevskaia et al., 2024]. One of the main reasons for the uneven distribution of the mitigation techniques is because of both technical and practical constraints. For example, In-processing approaches are more flexible because they can be implemented directly as constraints into the optimization objective function. In addition, there are more in-processing mitigation because of publication incentives: novel algorithmic contributions are often valued more than empirical comparisons of existing methods. Although this is the case, In-process mitigation techniques requires model retraining and are often architecture-specific. Post-processing methods are also popular because of easy implementation and they are not specific to any recommender model architecture [Klimashevskaia et al., 2024].

2.3 Pre-processing Methods

These methods tries to mitigate popularity bias by manipulating the input data in order to increase fairness in the recommender outputs. Pre-process mitigation are easy to implement as they do not involve tuning the recommender models. In the following subsection, we will discuss different types of Pre-process mitigation techniques.

Data Sampling and Rebalancing

The most direct pre-processing strategy involves altering the training data's composition through sampling. [Boratto et al., 2021] propose specific sampling strategies for both point-wise and pair-wise optimization settings. In their paper, they use popularity-aware sampling which reduces popularity bias by balancing training examples so the model learns equally from both popular and less popular items, rather than mostly favoring popular ones.

Item Exclusion and Catalog Partitioning

This approach involves removing or separating popular items from the training dataset in an attempt to reduce popularity bias. [Cremonesi et al., 2014] manipulates the training dataset by removing very popular items, this forces the model to learn and recommend less popular items too. [Park and Tuzhilin, 2008] propose a more specific way to exclude popular items from training data. Instead of removing the popular items, they first cluster the data into head and tail items. After that, the recommender model is then trained separately on the two clusters. The idea is to reduce the dominance nature of the popular items when head and long tail items are trained together. This way, the model is forced to also recommend unpopular items instead of only popular items.

Data Augmentation

Instead of removing popular items from the training data to reduce popularity bias, this approach creates a balance between popular and unpopular items in the dataset by data augmentation. Specifically, this is done by adding some additional information to the original dataset. For example, [Li et al., 2021] augments the training dataset by adding social connections information in order to help with sparsity interaction for long-tail items. [Jadidinejad et al., 2019] combine implicit and explicit feedback.

Summary

Pre-processing methods tries to mitigate popularity bias by manipulating the input data in order to increase fairness in the recommender outputs. Pre-process mitigation are easy to implement as they do not involve tuning the recommender models. However, it is important to make sure that data augmentation and sampling is done in such a way that the final datasets are still representative of the original datasets [Klimashevskaia et al., 2024].

2.4 In-processing Methods

In-processing methods try to mitigate popularity bias by including fairness constraints directly into the model training stage. In the following subsection, we will discuss different ways how In-processing methods are employed.

Regularization-based Approaches

The idea here is to penalize the model when it only recommends popular items. In so doing, this forces the model to also recommend unpopular items. For example, [Kamishima et al., 2014] employs regularization-based debiasing by adding a penalty term that measures the information between item popularity and recommendation scores in the training data. If the final recommender list has only higher scores for popular items, the model is then penalized. On the other hand, [Abdollahpouri et al., 2017] employs a learning-to-rank regularization framework that uses a LapDQ regularizer.

Constraint-based Approaches

Another way of reducing popularity bias is by including constraints in the optimization function of the models. For example [Wang and Wang, 2022] introduce a constraint in the objective function that reduces popularity bias by enforcing constraint that items which are similar should receive also similar coverage in final recommendations. Another way is proposed by [Seymen et al., 2021] where the recommendation decision objective function is extended by adding additional decision variables like auxiliary variables, or by defining upper and lower bounds, or including weighted sums.

Re-weighting Approaches

During model training, items are given weights based on their relevant. most recommender models tend to give high weights to popular items compared to unpopular items. One of the ways to mitigate popularity bias is through re-weighting approaches. For example, [Steck, 2011] proposes a method that tries to reduce the trade-off between accuracy and long-tail by either increasing the weights of for unpopular items or decreasing the weights of popular items.Another method is **Inverse Propensity Scoring (IPS)** which was proposed by [Schnabel et al., 2016]. This method manipulates the training weights of items by applying the inverse of the popularity-based propensity score.

Summary

In-processing methods tries to reduce popularity bias during the model training phase: There are several ways how this is done which includes regularization [Kamishima et al., 2014, Abdollahpouri et al., 2017], including constraints in the objective function [Wang and Wang, 2022, Seymen et al.,

2021], and re-weighting the scores of items [Steck, 2011,Zhao et al., 2013]. One main problem with In-processing mitigation strategies is that these methods usually require extensive hyperparameter tuning. Also, the optimal model parameter values obtained from one dataset might not generalize across a different datasets [Chen et al., 2023].

2.5 Post-processing Methods

Post-processing approaches are mitigation strategies that aims at reducing popularity bias by manipulating the recommender output. Here, the dataset and the base recommender model are not changed in anyway. This means that, the architecture of the recommender model used to generate the final recommender list does not need to be specific. The model-agnostic nature of Post-processing mitigation techniques gives them the practical advantages compared to In-process mitigation strategies. There are several techniques used to implement Post-processing approaches. Most of the techniques can be grouped into three groups; Re-ranking approaches, Re-scaling or Score Adjustment approaches, and Rank Aggregation approaches.

Re-ranking Approaches

Re-ranking mitigation strategies operates at the list level. The recommender model generates an initial sorted list that is later reordered by re-ranking approaches. The main idea for Re-ranking is to take the initial recommended list produced by a base recommender, reorder the item positions iteratively until a specific diversity or popularity constraint is met. There are several ways in which Re-ranking is implemented.

Calibrated Popularity

Proposed by [Abdollahpouri et al., 2021], Calibrated Popularity (CP) is a post-processing re-ranking method that selects a final recommendation list whose item-popularity distribution mirrors the popularity profile observed in the user's interaction history. Unlike item-centered approaches that treat all users uniformly, CP operates from a user-centric perspective: it acknowledges that different users have different tolerances toward popular contents, and that a fair recommendation should reflect those individual tendencies rather than imposing a globally uniform popularity target.

In Calibrated Popularity, Items are partitioned into three groups based on their popularity: head (H), mid-popularity (M), and tail (T). This

partitioning follows a Pareto-inspired split in which H items constitute the small fraction of items that account for roughly 20% of all interactions, T items are the large number of items that collectively account for another 20%, and M items cover the remainder.

For each user u, the historical popularity distribution is calculated over their interaction profile ρ_u:

$$p(c \mid u) = \frac{\sum_{i \in \rho_u} r_{u,i}\, \mathbf{1}[i \in c]}{\sum_{c' \in C} \sum_{i \in \rho_u} r_{u,i}\, \mathbf{1}[i \in c']}, \quad c \in C = \{H, M, T\}, \tag{2.1}$$

where:

- $r_{u,i}$ denotes the rating or implicit feedback of user u for item i.
- $\mathbf{1}[\cdot]$ is the indicator function

The corresponding popularity distribution over a candidate recommendation list L_u is:

$$q(c \mid u) = \frac{\sum_{i \in L_u} \mathbf{1}[i \in c]}{\sum_{c' \in C} \sum_{i \in L_u} \mathbf{1}[i \in c']}. \tag{2.2}$$

CP operates on an initial candidate pool of size m generated by a base recommender and selects a final list L_u^* of size $n \ll m$ by solving:

$$L_u^* = \arg \max_{L_u,\, |L_u|=n} \Big[(1-\lambda) \cdot \mathrm{Rel}(L_u) - \lambda \cdot J(P_u,\, Q(L_u)) \Big], \tag{2.3}$$

where:

- $\mathrm{Rel}(L_u)$ is the sum of predicted relevance scores for items in L_u.
- $J(P_u, Q(L_u))$ is the Jensen–Shannon divergence between the user's profile distribution $P_u = \{p(c \mid u)\}_{c \in C}$.
- the list distribution $Q(L_u) = \{q(c \mid u)\}_{c \in C}$, and $\lambda \in [0, 1]$ controls the trade-off between relevance and calibration.

The objective of Calibrated Popularity is to promote item relevance in a way that user's observed popularity tolerance is reflected in the final recommender list. The optimization is greedily solved using the maximum marginal relevance. This is done in steps; at each step, select an item that maximizes the objective, then move on to the second step, and repeat.

In their paper, [Abdollahpouri et al., 2021] using Last.fm and MovieLens 1M datasets, demonstrated that CP consistently achieves the lowest User Popularity Deviation (UPD). Although CP shows great performances, it also

has some limitations. First, the definitions of H/M/T partitions are not learned. Due to this, the H/M/T partition may not optimally reflect the popularity landscape of every domain. Second, due to its greedy search nature for L_u^*, this does not guaranty a globally optimal solution.

EXplicit Query Aspect Diversification (xQuAD)

[Abdollahpouri et al., 2019a] adapted the query diversification algorithm from [Santos et al., 2010] called xQuAD in order to use it as a re-ranking approach for popularity bias mitigation. This adaptation explicitly tries re-ranking the final recommender list also by balancing relevance and the marginal benefit of including under-exposed items. Like Calibrated Popularity [Abdollahpouri et al., 2021], xQuAD also defines the partitions of items into short-head and long-tail groups. Here, [Abdollahpouri et al., 2019a] defines the short-head as the few items that account for 80% of the total interaction. The main idea of xQuAD is that at each step of list construction, an item is selected by looking at both its predicted relevance score to the user and how much it contributes towards the popularity coverage objective of the final recommended list. Because each user has a historical ratio of head versus tail items, the item that needs to be promoted depends on this historical ration. In so doing, the final recommended list has a specific diversity target per user rather than being uniform.

[Abdollahpouri et al., 2019a] proposed two variants of EXplicit Query Aspect Diversification (xQuAD) which are **binary** xQuAD and **smooth** xQuAD. The main difference between the two variants is the way they treat long-tail membership. For **binary**, long-tail membership is treated as a hard indicator (Binary), which means that an item either belongs to the tail or it does not. On the other hand, **smooth** does not use a hard indicator, but rather uses continuous measure of "tailness" by using the inverse popularity of an item to weight its contribution proportionally. This provides a finer-grained way to promote less popular items into the final recommendation list.

One limitation of xQuAD is that it only uses two partition groups of item interactions; short-head and long-tail instead of three groups like in [Abdollahpouri et al., 2021]. In the end, this can collapse the richer three-group structure (head, mid, tail) which can help to provide better ways to mitigate popularity bias. In this paper, both variants are adopted as baselines.

FA*IR

FA*IR [Zehlike et al., 2017] proposed a mitigation strategy that specifically aims at protecting under-represented groups in recommendation list. The main focus of FA*IR Defines what should be the minimum proportion of items that belong to the protected group. This minimum proportion of protected is checked at every level of the item ranking. The definition of protected ground varies based on the objective of the fairness. Some common examples of protected candidates may have attributes belonging to a specific ethnic group or gender. In our paper we defined the protected group base on popularity bias. Specifically, we first define short-head and long-tail groups like [Abdollahpouri et al., 2019a], then long-tail items are treated as the protected group. In our case FA*IR is used to provide a positional guarantee on the representation of less popular items at every level of the recommendation list.

However, FA*IR has a limitation that it tries to promote fairness on group-level and not on user-level. Although globally, FA*IR is able to promote fairness, it can also hinder the goal of creating personalized recommendation list. In this mitigation strategy is also adopted in this paper.

FairMatch

Another interesting approach to performing re-ranking to a recommended list is by using Graph-based re-ranking methods [Mansoury et al., 2020a]. Specifically, the method uses network structure to find relevant items that are not popular and tries to promote them up the ranked list. This improves visibility of these high-quality items that are rarely recommended. FairMatch is different from other mitigation strategies because it operates on system level compared to the other mitigation strategies rather operate on group or user level. Because of its system-level nature, FairMatch is able to affect both the user side and supplier side of the recommender system.

FairMatch uses a bipartite graph to connect users and items so that it can identify these high-quality items that have low-visibility. Earlier graph-based approaches tried to impose a fixed target distribution in order to solve the recommendation problem [Antikacioglu and Ravi, 2017]. This is one of the main improvement of FairMatch over the old methods. From the start, FairMatch does not use any assumption or have a fixed distribution target but instead, it lets the flow structure of the graph map out the distribution of items that are underexposed relative to their relevance. This gives the algorithm more freedom to promote genuinely high-quality long-tail items without being constrained by a pre-specified target.

FairMatch works as a batch re-ranking process. After the recommender

base model has generated recommendation for each user, FairMatch refines the list through iterative maximum flow problem solutions that are applied to a recommendation bipartite graph. specifically, this process identifies high-quality, low-visibility item subgraphs, extracting and removing them from the graph at each iteration until a final recommendation list of the desired size is created. The graph's edge weights represent item visibility (node degree) and relevance (rank position), with a hyperparameter α balancing these factors.

One of the main limitations of FairMatch is that it treats all users uniformly when it is making the fairness adjustments. this in the ends reduce the personalization of the final recommendation list. Secondly, FairMatch has a high computational complexity compared to single-pass re-ranking approaches. specifically, it has an upper bound computational complexity of $\mathcal{O}(V^3E)$, where V and E denote the number of nodes and edges in the graph.

Re-scaling (Score Adjustment)

Re-scaling operates at the score level. Unlike Re-ranking approaches, where they work on the initial sorted list from recommender model, here the recommender model only generates raw scores. Re-scaling then modifies these raw preference scores that were generated by the recommendation model before the final recommendation list is sorted. Specifically, Re-scaling approach uses the popularity score of an item in order to apply a mathematical penalty to that item. If an item is popular, its score is reduced and if the item is a long-tail item, then its score is not changed or it is increased. After the scores have been re-scaled, then a final list is generated.

Popularity Compensation

[Zhao et al., 2022] proposes that in order to reduce popularity bias, a compensation score should be added to the initial predicted scores for unpopular items so that they are promoted. On the other hand, the score for popular items should be reduced in order to reduce their dominant effect.

Let $R \in \{0,1\}^{N\times M}$ be the user–item interaction matrix, where $R_{u,i} = 1$ if user u interacted with item i in training and 0 otherwise.

Let $\hat{R} \in \mathbb{R}^{N\times M}$ be the predicted score matrix produced by a base CF model (e.g., MF or BPR).

For user u, denote by $O_u^+ = \{i \mid R_{u,i} = 1\}$ the set of items interacted with by user u in training. The number of items is M, and the number of items in O_u^+ is $|O_u^+|$.

Only items that user u has not interacted with in training are considered as candidates, i.e., items with $R_{u,i} = 0$.

For user u, define the scale of the original scores over candidate items as

$$n_u = \left\| \frac{\hat{R}_{u,:} \odot (\mathbf{1} - R_{u,:})}{M - |O_u^+|} \right\|_F,$$

where $\odot$ denotes elementwise (Hadamard) product, $\mathbf{1}$ is the all-ones vector, and $\| \cdot \|_F$ is the Frobenius norm.

Let $\text{pop}(i)$ denote the popularity of item i, typically measured as the number of users that interacted with item i in the training data.

For a given user u and item i, define the (unnormalized) compensation term

$$C_{u,i} = \frac{1}{\text{pop}(i)} \cdot \big(\beta \, \hat{R}_{u,i} + (1 - \beta)\big),$$

where $\beta \in [0, 1]$ controls the degree of preference-awareness:

- $\beta = 0$: compensation depends only on item popularity;
- $\beta = 1$: compensation is proportional to the base score $\hat{R}_{u,i}$.

Define the scale of the raw compensation for user u as

$$m_u = \left\| \frac{C_{u,:} \odot (\mathbf{1} - R_{u,:})}{M - |O_u^+|} \right\|_F,$$

where $C_{u,:}$ denotes the vector of $C_{u,i}$ over all items i.

The debiased (post-processed) score for user u and item i is

$$\hat{R}^*_{u,i} = \hat{R}_{u,i} + \alpha \, C_{u,i} \, \frac{n_u}{m_u},$$

where $\alpha \geq 0$ is a hyperparameter that controls the overall strength of debiasing versus preserving the original scores.

At recommendation time, items are ranked for each user u according to $\hat{R}^*_{u,:}$ instead of $\hat{R}_{u,:}$.

One limitation of Popularity Compensation mitigation strategy is that the method is tuned with hyperparameter, so the result depends on choosing good values for α and β

Rank Aggregation

Rank Aggregation operates at the multi-list level. Here there is an assumption that a recommender base model is used to generate multiple distinct ranked lists for a single user. Then these lists are merged to create one final recommendation list that has a fairness goal optimized. For example, two separate

recommender list can be generated, first one being generated by a model that was optimized purely on accuracy and then other list being generated from a model that was optimized for only fairness. In so doing, there will be two list that are essential different. After that, a Rank Aggregation method that uses either permutation consensus methods (like Borda Count, Kemeny-Young, or Markov Chain methods) or voting is used to combine the two lists to form one final list. Specifically, the Rank aggregation method, looks at the two lists as voters and tries to find the best items that performs well by promoting the long-tail objectives and and not hurting accuracy so much. The long-tail objectives differ from one method to another.

Two-Way Rank Aggregation (TWRA)

[Dong et al., 2019] propose the Two-Way Rank Aggregation (TWRA) a method that integrates two distinct ranking approaches:

- **Forward rank**: It focuses on user preferences by sorting uncollected items based on predicted relevance.
- **Backward rank**: It organizes items according to predicted relevance scores among users who have yet to interact with them.

The final aggregated ranking is produced by a weighted linear combination of the two rank numbers:

$$\text{ag-rank}_\pi(u, i) = (1 - \lambda) \cdot \text{f-rank}_\pi(u, i) + \lambda \cdot \text{b-rank}_\pi(u, i) \tag{2.4}$$

where

- $\lambda \in [0, 1]$ controls the relative weight of the item-oriented perspective.

When $\lambda = 0$ the framework reduces to the original base algorithm; when $\lambda = 1$ the ranking is determined entirely from the item's perspective, disregarding user preference.

On MovieLens and Netflix datasets, TWRA demonstrates significant enhancements in diversity (measured by Hamming Distance) and coverage (Gini coefficient) compared to the base model and its variations. It is able to maintaining accuracy or in some cases improving accuracy, especially in sparser datasets where the backward rank adds essential information.

One limitation of TWRA is that it aggregates ranks instead of raw scores, neglecting the value of score magnitude differences. Additionally, TWRA employs a uniform λ for all users, lacking personalization for individual user preferences regarding popular items.

Summary

Post-processing methods adjust already-generated recommendation lists via re-scaling [Zhao et al., 2022], re-ranking [Abdollahpouri et al., 2021, Klimashevskaia et al., 2022], and rank aggregation [Dong et al., 2019, Yalcin and Bilge, 2021]. The primary advantage of post process is that it is independent from the underlying model as it only rearrange existing candidates.

2.6 Fairness-Utility Trade-off in Debiasing

Majority of literature on popularity bias mitigation frequently evaluates recommendation quality using both accuracy and beyond-accuracy metrics. On the accuracy side, studies commonly report ranking measures such as NDCG and Recall, while on the bias or exposure side they often use popularity-oriented measures such as Average Recommendation Popularity (ARP) and long-tail exposure indicators such as Average Percentage of Long-Tail items (APLT) [Abdollahpouri et al., 2017]. These metrics are particularly useful because they capture whether a mitigation strategy reduces concentration on popular items and increases exposure of niche or long-tail items, making them standard choices for analyzing the accuracy–fairness trade-off in recommender systems.

2.7 Chapter Summary

The model-agnostic nature of post-processing methods makes them easy to implement, and prior work shows that they can effectively reduce popularity bias. However, the literature also reveals a significant gap regarding when and how these methods remain effective. Because post-processing operates only on an initial candidate list which is generated by the base recommender model, its impact is fundamentally dependent on the initial composition of that list. In other words, the debiasing effectiveness of a post-processing method depends essentially on the severity and structure of the base model's bias, which may vary so much across algorithms used by the base recommender and architectures. This dependency is rarely examined explicitly, leaving open the question of how robust post-processing strategies are when the underlying model or its popularity bias profile changes. Addressing this gap directly motivates the experimental design and cross-architecture validation framework presented in Chapter 3

Chapter 3

Methodology

This chapter presents a controlled experimental framework designed to systematically investigate the sensitivity of fairness-aware reranking methods to varying degrees of popularity bias. While prior work typically evaluates fairness interventions on static, naturally occurring datasets, such approaches do not allow for causal analysis of how mitigation effectiveness changes as bias intensifies. To address this limitation, we introduce a controlled bias induction procedure that enables fine-grained manipulation of the item popularity distribution, while preserving realistic interaction patterns.

3.1 Research Questions and Hypotheses

The research questions and hypotheses introduced in chapter 1 guide the design of the experiments in this chapter:

- **RQ1: Measure Bias Impact** — How do different levels of popularity bias in the base recommender affect the performance of post-processing mitigation methods?
- **RQ2: Map Trade-offs** — How do fairness–utility trade-offs change as bias increases?
- **RQ3: Compare Methods** — How robust are different post-processing strategies under rising bias across datasets and model types?

We try to answer these questions through the two-stage experimental framework described in the following sections, which systematically varies the popularity bias level α and evaluates the performance of fairness-aware reranking methods.

Building on these questions, the study is guided by the following hypotheses (first stated in Chapter 1):

- **H1:** The effectiveness of fairness-aware reranking methods decreases as popularity bias α increases.
- **H2:** Different reranking strategies exhibit varying robustness to increasing bias levels.
- **H3:** The optimal fairness–utility trade-off parameter is dependent on the underlying bias distribution.

In the subsequent sections we describe how each hypothesis is tested through the experimental design, including bias induction, candidate generation, reranking, and multi-objective optimization.

3.2 Experimental Framework

Two-Stage Architecture

To isolate the effect of fairness interventions, we adopt a modular two-stage recommendation pipeline. This design decouples the learning dynamics of the base recommender from the fairness-aware post-processing step, thereby enabling a cleaner causal interpretation of reranking effectiveness as a function of base-model bias.

- **Stage 1: Candidate Generation.** A collaborative filtering model is trained on a biased dataset. For each user u, the model produces a ranked candidate set $\mathcal{C}_u$ of the top $N = 100$ items based on predicted relevance scores.
- **Stage 2: Fairness-Aware Reranking.** A reranking algorithm reorders $\mathcal{C}_u$ to produce a final top-K list ($K = 10$), optimizing for increased long-tail exposure while preserving ranking utility.

This two-stage architecture directly supports RQ1 and Hypothesis H1, as it allows us to vary the popularity bias level α in Stage 1 while keeping the reranking methods in Stage 2 fixed, thus isolating the effect of base-model bias on mitigation effectiveness.

Fairness Definition We define fairness as *provider-side exposure fairness*, where the objective is to ensure that items from the long-tail receive equitable visibility relative to their presence in the catalog [Abdollahpouri et al., 2017].

3.3 Dataset Description

In order to investigate how the increase in popularity bias in the dataset, affects the ability of a post-processing mitigation method to promote long tail item coverage while preserving recommendation accuracy, we adopt a controlled experimental approach in which the level of bias in the datasets is systematically varied. We used two datasets MovieLens-1M [Harper and Konstan, 2015] and Amazon Books [Ni et al., 2019].

Original Dataset: MovieLens-1M

The MovieLens-1M dataset is a widely adopted benchmark dataset in recommender systems research. It contains 1,000,209 ratings (1–5) from 6,040 users on 3,952 movies, collected by the GroupLens research group. Each user has rated at least 20 movies [Harper and Konstan, 2015]. The dataset exhibits natural long-tail characteristics: a small fraction of items account for a large proportion of interactions.

Original Dataset: Amazon Books

The Amazon Books dataset is a widely adopted benchmark dataset in recommender systems research. It contains user–book ratings collected from Amazon, and we use the *ratings-only* version, which includes only $(user, item, rating, timestamp)$ tuples and no review text or metadata. The Books ratings-only subset contains 22,507,155 ratings [Ni et al., 2019] . Similar to MovieLens and other real-world recommendation datasets, it exhibits strong sparsity and long-tail characteristics. We apply the same iterative 20-core filter, which requires 32 passes due to the dataset's extreme sparsity. After filtering, the dataset contains **35,736 users**, **38,121 items**, and **1,960,674 interactions** [Abdollahpouri et al., 2019a].

Bias Induction via Power Law Interaction Resampling

To systematically control the level of popularity bias in MovieLens and Amazon Books datasets, we construct a family of training datasets via power-law interaction resampling. The power-law form is a natural choice given that item interaction frequencies in recommendation datasets exhibit heavy-tailed structure [Abdollahpouri et al., 2017].

First, we computed the popularity of each item in the dataset. Let p_i denote the empirical popularity of item i in the filtered source data, defined

as the proportion of interactions associated with item i [Lin et al., 2025]:

$$p_i = \frac{R_i}{R_{\text{total}}} \tag{3.1}$$

where:

- R_i is the number of interactions for item i in the filtered source data
- R_{total} is the total number of interactions

For a given exponent $\alpha \geq 1$, we assign each interaction involving item i a sampling weight

$$w_i \propto p_i^{\alpha-1} \tag{3.2}$$

and draw a fixed number of interactions *with replacement* according to these weights [Gan and Jiang, 2013]. The exponent α directly controls the degree of popularity amplification: when $\alpha = 1.0$, all weights are equal ($w_i \propto 1$) and the sampled dataset preserves the natural popularity distribution of the source data; for $\alpha > 1.0$, the interaction mass becomes progressively concentrated among fewer head items, producing increasingly head-dominant training sets.

We generate five bias levels: $\alpha \in \{1.0, 1.2, 1.5, 1.75, 2.0\}$. Target interaction counts were **500,000** for MovieLens-1M (50.2% of the original) and **750,000** for Amazon Books (38.3%). Three independent random seeds $\{42, 123, 456\}$ are used at each $(\text{dataset}, \alpha)$ condition, yielding $5 \times 3 \times 2 = 30$ training runs for the BPR sensitivity analysis.This matched-condition design enables us to evaluate how base models trained on different bias levels respond to fairness interventions, with each α representing a distinct operational scenario.

Choice of sampling with replacement.

The use of replacement is necessary for true power-law amplification at higher α values. Without replacement, the maximum count for any item is bounded by its original frequency in the source data, which suppresses the desired concentration effect, particularly at $\alpha = 2.0$. While sampling with replacement introduces duplicate interactions and may distort the natural user–item interaction process, this design choice is intentional. The objective of this study is not to faithfully reproduce real-world interaction dynamics, but to construct controlled datasets in which popularity bias can be systematically varied as a single experimental factor. Under this controlled setting, replacement-based sampling provides a simple and effective mechanism to generate datasets with progressively increasing skew, enabling consistent and interpretable analysis of model behavior under varying levels of popularity bias.

Fixed Head–Tail Partitioning

To facilitate analysis of model performance across the popularity spectrum, we partition items into head and tail segments using a cumulative interaction mass threshold, following the convention established by [Abdollahpouri et al., 2017]. Specifically, items are ranked by their number of interactions in descending order (3.1), and the head set is defined as the minimal set of items whose cumulative interactions account for 80% of all ratings in the biased training sample. All remaining items constitute the tail.

A critical methodological decision is that this partition is computed *once* from the $\alpha = 1.0$ distribution and held **fixed** across all bias levels. Using original item identifiers, the same head–tail boundary is applied at every α condition. This ensures that the fairness metric APLT@10 always measures exposure to the same set of genuinely niche items, enabling meaningful cross-α comparison.

The resulting fixed partition sizes are:

- **MovieLens-1M:** 1,167 head items (38.4%), 1,876 tail items (61.6%).
- **Amazon Books:** 22,103 head items (58.0%), 16,018 tail items (42.0%).

As α increases, some tail items accumulate zero interactions in the biased dataset and disappear from model training entirely. These structurally missing items cannot receive meaningful embeddings and therefore cannot appear in any user's candidate pool regardless of the reranking method applied. This structural depletion is the mechanism behind the declining effectiveness of post-processing at high bias levels..

Tail Remapping

Because resampling with replacement may exclude certain items from specific biased datasets, leading to index inconsistencies between conditions, the long-tail set is tracked using original item identifiers and remapped to internal model indices at each experimental condition. Items absent from training at a given α level are recorded as structurally missing and are handled appropriately during evaluation.

3.4 Stage 1: Candidate Generation

Base Model: Bayesian Personalized Ranking (BPR)

Bayesian Personalized Ranking is a pairwise ranking optimization framework for implicit feedback. Unlike pointwise methods that predict absolute ratings,

BPR optimizes for the correct relative ordering of observed versus unobserved items.

Optimisation objective. BPR maximises the posterior probability over training triplets (u, i, j), where user u has interacted with item i but not with item j:

$$\text{BPR-OPT} = \sum_{(u,i,j)\in D_S} \ln \sigma(\hat{x}_{ui} - \hat{x}_{uj}) - \lambda_\Theta \|\Theta\|^2 \tag{3.3}$$

where:

- D_S denotes the set of training triplets,
- σ is the logistic sigmoid function,
- λ_Θ is the regularisation hyperparameter.

Model Architecture. We implement BPR with matrix factorization using $d = 64$ latent dimensions. The prediction for user u and item i is:

$$\hat{x}_{ui} = \mathbf{p}_u^\top \mathbf{q}_i \tag{3.4}$$

where:

- $\mathbf{p}_u \in \mathbb{R}^d$ and $\mathbf{q}_i \in \mathbb{R}^d$ are user and item embeddings,
- d denotes the dimensionality of the embedding space.

Training configuration:

- Embedding dimension: $d = 64$
- Epochs: 50
- Optimiser: AdamW (weight decay = 0.01)
- Learning rate: 0.02
- Batch size: 2,048
- Loss: BPR pairwise ranking loss
- Negative sampling: uniform random from unobserved items ($\leq$ 100 attempts per sample)

Candidate generation. After training, the model computes predicted scores for all user–item pairs. Training items are excluded (score set to $-\infty$) and the top-100 unobserved items by predicted score are retained as the candidate pool $\mathcal{C}_u$ for each user. These candidate lists, together with user interaction profiles and evaluation bundles, are saved to disk before Stage 2 begins, ensuring that all mitigation methods operate on identical inputs.

Bias Characterization at Candidate Generation Stage

To characterise the bias present in the base model's output, we compute two complementary measures.

Average Recommendation Popularity (ARP@K). Following [Abdollahpouri et al., 2017], ARP measures the degree to which the model favours popular items:

$$\text{ARP@}K = \frac{1}{|U|} \sum_{u \in U} \frac{1}{K} \sum_{i \in R_u} \text{pop}(i) \tag{3.5}$$

where:

- R_u is the recommendation list for user u
- $\text{pop}(i)$ is the interaction count of item i in the training set.

Pool long-tail fraction. We additionally compute the pool long-tail fraction as the mean percentage of fixed-definition tail items in each user's top-100 candidate pool, averaged across all users and seeds:

$$\text{Pool-LT} = \frac{1}{|U|} \sum_{u \in U} \frac{|\{i \in \mathcal{C}_u : i \in L\}|}{N} \tag{3.6}$$

where:

- L is the fixed long-tail set
- $N = 100$

This metric quantifies the structural upper bound on long-tail promotion: no post-processing method can recommend a tail item that does not appear in the candidate pool.

Evaluation Metrics

The performance of the base model was evaluated on two types of metrics. First one is for accuracy family and the other is fairness metrics. Each serves a specific purpose in order to track all relevant information on the performance of the model.

Accuracy Metrics

- **NDCG@10.** Normalised Discounted Cumulative Gain measures ranking quality by giving higher credit to relevant items appearing earlier:

$$\text{NDCG@10} = \frac{\text{DCG@10}}{\text{IDCG@10}}, \qquad \text{DCG@10} = \sum_{p=1}^{10} \frac{1[\text{item}_p \text{ relevant}]}{\log_2(p+1)} \tag{3.7}$$

  where IDCG is the DCG of an ideal ranking with all relevant tems at the top and NDCG is averaged over all test users.

- **Recall@10.** The fraction of relevant items (those in the test set) that appear in the top-10 recommendations:

$$\text{Recall@10} = \frac{|\{\text{recommended items}\} \cap \text{test items}_u|}{\min(10,\ |\text{test items}_u|)} \tag{3.8}$$

Fairness Metrics

- **Average Percentage of Long-Tail items (APLT@10).** The proportion of recommended items that belong to the fixed long-tail set L [Abdollahpouri et al., 2017]:

$$\text{APLT@10} = \frac{1}{|U|} \sum_{u \in U} \frac{|\{i \in R_u : i \in L\}|}{10} \tag{3.9}$$

  A higher APLT indicates greater exposure of niche items. Because L is fixed across all α conditions, this metric is directly comparable across bias levels.

- **Average Recommendation Popularity (ARP@10).**

$$\text{ARP@10} = \frac{1}{|U|} \sum_{u \in U} \frac{\sum_{i \in R_u} \text{pop}(i)}{10} \tag{3.10}$$

  Lower ARP indicates that recommended items are on average less popular.

3.5 Stage 2: Fairness-Aware Reranking

We evaluate three post-processing reranking algorithms designed to increase the exposure of long-tail items while preserving recommendation quality. All methods receive as input the top-100 candidates from Stage 1 with their predicted relevance scores, the user's training interaction history, and the fixed long-tail item set L.

EXplicit Query Aspect Diversification(xQuAD)

The original xQuAD [Santos et al., 2010] was adopted to create a xQuAD framework to balance relevance and exposure diversity by [Abdollahpouri et al., 2019a]. At each step, an item is selected according to:

$$(1-\lambda)\,P(i \mid u) + \lambda \sum_{c \in \{\text{head},\,\text{tail}\}} P(c \mid u)\,P(i \mid c) \prod_{j \in S} (1 - P(j \mid c)) \tag{3.11}$$

We evaluate two variants following [Abdollahpouri et al., 2019a]:

- **Binary xQuAD:** Hard head/tail partitioning.
- **Smooth xQuAD:** Continuous popularity-based weighting

Binary xQuAD

Binary xQuAD treats long-tail items as a "sub-group" that should be promoted. Given a user u and a ranked list of candidate items C, the method incrementally builds a reranked list R. At each step, the next item i is selected according to:

$$\lambda \cdot \text{rel}(i) + (1-\lambda) \cdot \text{cov}(i) \tag{3.12}$$

where:

- $\text{rel}(i)$ is the relevance score from the base model,
- $\text{cov}(i)$ measures coverage of uncovered aspects (long-tail items),
- $\lambda \in [0, 1]$ controls the trade-off between accuracy and fairness.

In the binary variant, an item either belongs to the long-tail set or not; the fairness term is simply an indicator of whether i is a long-tail item, normalized by the number of long-tail items already selected. The parameter $\lambda \in [0, 1]$ controls the trade-off between relevance and fairness. When $\lambda = 0$, the original ranking is preserved; when $\lambda = 1$, only fairness (long-tail promotion) matters.

Smooth xQuAD

Smooth xQuAD replaces the binary grouping with a continuous measure of "tailness". Instead of a hard partition, each item is assigned a weight w_i proportional to its inverse popularity (e.g., $w_i = 1/\text{popularity}(i)$).

$$\lambda \cdot \text{rel}(i) + (1 - \lambda) \cdot \frac{\text{rel}(i) \cdot \text{cov}(i)}{\max(\text{rel}) \cdot \max(\text{cov})} \tag{3.13}$$

The fairness term then becomes the sum of weights of long-tail items already in the list, and the contribution of candidate i is the marginal gain in total weight if i were added. This formulation avoids abrupt boundaries and can be more appropriate when popularity is a spectrum rather than a dichotomy. The same λ parameter governs the trade-off.

FA*IR

FA*IR (Fairness-Aware Reranking) originates from the fair ranking literature [Zehlike et al., 2017] and enforces statistical parity constraints. It requires that in the top-k recommendations, the proportion of protected items (here, long-tail items) meets a minimum threshold. The method operates as follows:

Phase 1 — Minimum Selection. For each rank position t, compute the minimum number of long-tail items that must appear among the top t positions to satisfy a given proportion p with statistical significance level α (typically 0.05). Formally, the minimum threshold at position t is:

$$m_t = \max\big\{m \in \{0, \ldots, t\} : P(X < m) \leq \alpha_s, \quad X \sim \text{Binomial}(t, \tau)\big\} \tag{3.14}$$

Phase 2 — Constrained Reranking. Construct the final ranking by iteratively selecting the highest-relevance item that satisfies the fairness constraint at each position.

Formally, FA*IR ensures that after k selections, the number of protected items is at least $\lceil \alpha \cdot k \rceil$ with probability controlled by a statistical test. In our implementation, we use a deterministic version: at each step, if the current proportion of protected items is less than the target p, we select the highest-scoring protected candidate; otherwise, we select the highest-scoring overall candidate. This guarantees the parity constraint exactly. The parameter τ (denoted p in the original paper) is the minimum desired fraction of long-tail items in the top-k. We sweep τ from 0.1 to 0.9 in steps of 0.1.

3.6 Parameter Selection via Nash Bargaining

Rather than reporting results across all parameter values, we use Nash bargaining to identify a single representative operating point per method that balances accuracy and fairness. For each parameter value λ (or τ for FA*IR), the Nash score is defined as:

$$\text{Nash}(\lambda) = \frac{\text{NDCG}(\lambda)}{\text{NDCG}_{\text{base}}} \times \frac{\text{APLT}(\lambda)}{\text{APLT}_{\text{base}}} \tag{3.15}$$

where:

- $\text{NDCG}_{\text{base}}$ and $\text{APLT}_{\text{base}}$ are the unmitigated BPR baseline values at the corresponding α level.

This multiplicative formulation of relative improvements ensures Pareto efficiency, symmetry, and scale invariance [Nash et al., 1950, Caragiannis et al., 2019]: a parameter that severely reduces NDCG will produce a low Nash score even if it maximises APLT, naturally penalising extreme operating points.

Nash scores are computed for each $(\alpha, \text{method}, \lambda)$ combination and averaged across all five α levels to identify a single *cross-α consensus* parameter. This consensus parameter is used without modification for the cross-model validation experiments, enabling direct comparison of mitigation effectiveness across architectures under matched conditions.

3.7 Experimental Procedure

Stage 1: Candidate Generation

For each combination of dataset $\in \{\text{MovieLens-1M}, \text{Amazon Books}\}$, bias level $\alpha \in \{1.0, 1.2, 1.5, 1.75, 2.0\}$, and seed $\in \{42, 123, 456\}$:

1. **Data loading.** The biased training and test sets for the given α are loaded from disk.

2. **Long-tail remapping.** The fixed long-tail set (computed from the $\alpha = 1.0$ distribution using original item identifiers) is remapped to the internal item indices used by the current model. Items absent from training at this α level are recorded as structurally missing.

3. **Model training.** A BPR model is trained for 50 epochs using Adam with the configuration in Section 3.4. The same hyperparameters are used across all α conditions to ensure comparability.

4. **Candidate generation.** For each test user, the top-100 items (excluding training interactions) are retrieved by predicted relevance score.

5. **Baseline evaluation.** The candidate list truncated to top-10 serves as the unmitigated baseline against which all metrics are computed.

6. **Persistence.** Candidate lists, user profiles, and evaluation bundles are saved to disk. This decouples Stage 1 from Stage 2, enabling mitigation methods to be rerun independently without repeating model training.

Stage 2: Mitigation Sweep and Parameter Selection

For each saved candidate set (dataset $\times$ α $\times$ seed), all three mitigation methods are applied across their respective parameter grids:

- Binary xQuAD and Smooth xQuAD: $\lambda \in \{0.0, 0.1, 0.2, \ldots, 1.0\}$
- FA*IR: $\tau \in \{0.1, 0.2, \ldots, 0.9\}$

For each parameter value, the reranking algorithm reorders the candidate list to produce a top-10 recommendation and all four metrics (NDCG@10, Recall@10, APLT@10, ARP@10) are computed against the test set. Results are aggregated across seeds and saved per $(\alpha, \text{method}, \lambda)$ combination.

Nash scores are then computed as described in Section 3.6. The parameter maximising the mean Nash score across all α levels is selected as the cross-α consensus and stored for subsequent cross-model validation.

Cross-Model Validation

To assess whether the sensitivity pattern and Nash-selected parameters generalise beyond BPR, we validate on two additional collaborative filtering architectures:

- **LightGCN** [He et al., 2020]: A simplified graph convolutional network that removes nonlinear transformations and feature projection, retaining only neighbourhood aggregation. Trained for 50 epochs with 3 propagation layers, embedding dimension 64, learning rate 0.02, and AdamW optimiser.
- **NeuMF** [He et al., 2017]: Neural Matrix Factorisation combining generalised matrix factorisation and a multi-layer perceptron. Trained for 50 epochs with MLP hidden dimensions $[64, 32, 16, 8]$, dropout rate 0.2, learning rate 0.02, and AdamW optimiser.

Both models are trained from scratch for each (dataset, α, seed) combination using the same biased training datasets as BPR, yielding $5 \times 3 \times 2 \times 2 = 60$ additional training runs. The Nash-selected parameters are applied without modification to all architectures, enabling direct comparison of mitigation effectiveness across models under matched conditions. As in Stage 1, all candidate lists are saved to disk prior to reranking, so that mitigation results can be reproduced without retraining the base models.

This methodology ensures a rigorous, reproducible evaluation of the three mitigation strategies under controlled popularity bias, enabling clear conclusions about their effectiveness and the optimal tuning required as bias intensifies.

Chapter 4

Experiments and Results

This chapter presents the results of the two-stage experimental framework introduced in Chapter 3. Section 4.1 validates the popularity-bias induction procedure and examines the resulting candidate-pool characteristics in Stage 1. Section 4.2 reports the performance of the unmitigated BPR recommender. Section 4.3 introduces the Nash-based parameter selection process, while Section 4.4 evaluates mitigation effectiveness and identifies the resulting structural ceiling. Section 4.5 investigates the zone-migration mechanism underlying the ceiling behavior. Section 4.6 analyzes the accuracy–fairness trade-off, and Section 4.7 extends the evaluation across multiple recommendation architectures. The implications of these findings for the research questions and hypotheses are discussed in Chapter 5.

4.1 Bias Induction and Candidate-Pool Composition

This section validates the bias induction procedure of Stage 1 and characterizes its consequences for the candidate pool from which all post-processing methods must draw. The analysis proceeds in three steps: first, we confirm that the resampling parameter α produces a controlled, monotone gradient of popularity bias (Section 4.1); second, we show that this gradient systematically depletes the long-tail content of each user's top-100 candidate pool (Section 4.1); and finally, we examine *where* within the pool the surviving long-tail items are located (Section 4.1). Together, these results establish the supply-side constraint against which all subsequent mitigation results must be interpreted.

Bias Statistics Across α Levels

Tables 4.1 and 4.2 report the key statistics of the datasets created by bias induction at each α level. Because the long-tail partition is fixed using the $\alpha = 1.0$ distribution, tail coverage measures how many of the originally defined tail items retain at least one training interaction as α increases.

Table 4.1: Bias statistics for MovieLens-1M across α levels.

α	Gini	ARP	Tail coverage	Missing tail items
1.00	0.562	819.5	100.0%	0
1.20	0.636	930.2	100.0%	0
1.50	0.720	1092.7	98.7%	24
1.75	0.767	1225.2	92.1%	148
2.00	0.802	1353.5	83.5%	310

Table 4.2: Bias statistics for Amazon Books across α levels.

α	Gini	ARP	Tail coverage	Missing tail items
1.00	0.412	112.7	≈100%	13
1.20	0.496	139.2	99.9%	66
1.50	0.619	191.8	99.1%	337
1.75	0.710	250.4	95.3%	1,234
2.00	0.784	320.9	85.8%	3,121

All bias indicators exhibit a monotonic response to increasing α on both datasets. On MovieLens-1M, the Gini coefficient of the interaction distribution increases from 0.562 to 0.802, while on Amazon Books it increases from 0.412 to 0.784. Consistent with this trend, the average recommendation popularity (ARP) increases and tail coverage decreases as bias intensifies. The Spearman rank correlation between α and the Gini coefficient is $+1.000$ ($p < 0.001$) for both datasets, confirming a perfectly monotonic relationship. These results demonstrate that the resampling procedure generates a stable and controllable gradient of popularity bias, validating α as an effective experimental control variable.

At the most extreme bias level ($\alpha = 2.0$), 310 of the 1,876 long-tail items in MovieLens-1M and 3,121 of the 16,018 long-tail items in Amazon Books are completely absent from the training data. This provides an early indication that severe popularity bias does more than re-weight item frequencies: it

removes portions of the catalogue from the model's observable training space altogether, limiting their opportunity to be recommended.

Effect on Candidate-Pool Composition

Figures 4.1 and 4.2 shows the Gini coefficient (panel a) and the candidate-pool long-tail fraction (panel b) as functions of α. The pool long-tail fraction is the mean percentage of fixed-definition tail items in each user's top-100 candidate pool, averaged over all users and three seeds; Table 4.3 reports the underlying values.

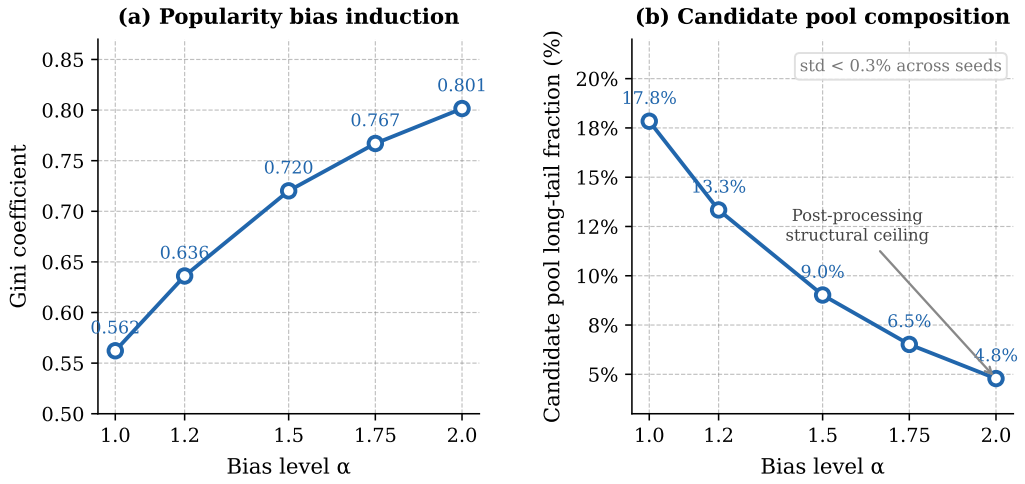


Figure 4.1: MovieLens-1M: (a) Gini coefficient vs. α, confirming monotone bias induction. (b) Candidate-pool long-tail fraction vs. α, establishing the structural ceiling on post-processing effectiveness.

Table 4.3: Candidate-pool long-tail fraction (mean ± std over 3 seeds).

α	MovieLens-1M	Amazon Books
1.00	$17.8\% \pm 0.1\%$	$15.1\% \pm 0.2\%$
1.20	$13.3\% \pm 0.3\%$	$11.9\% \pm 0.2\%$
1.50	$9.0\% \pm 0.1\%$	$8.3\% \pm 0.2\%$
1.75	$6.5\% \pm 0.1\%$	$6.2\% \pm 0.1\%$
2.00	$4.8\% \pm 0.1\%$	$4.5\% \pm 0.2\%$

On MovieLens-1M, the proportion of long-tail items in the candidate pool decreases from 17.8% at $\alpha = 1.0$ to 4.8% at $\alpha = 2.0$. Amazon Books exhibits

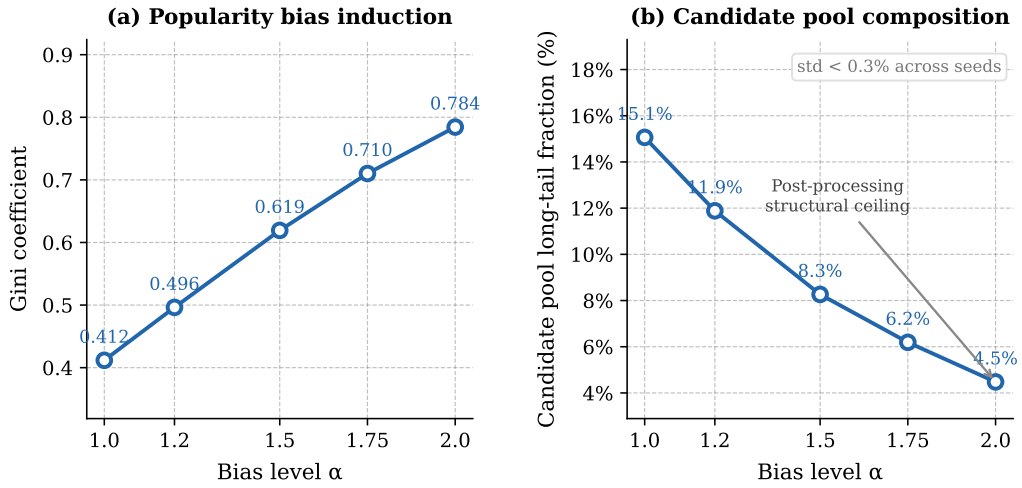


Figure 4.2: Amazon Books: (a) Gini coefficient vs. α, confirming monotone bias induction. (b) Candidate-pool long-tail fraction vs. α.

the same behavior, with the long-tail fraction declining from 15.1% to 4.5% (Figure 4.2b). Across all bias levels, the standard deviation over random seeds remains below 0.3%, indicating that the observed decline is highly consistent. This demonstrates that candidate-pool depletion is a stable consequence of increasing popularity bias rather than an artifact of random initialization.

The pool long-tail fraction provides the critical connection between popularity bias and mitigation effectiveness. By construction, post-processing methods can only rerank items that are already present in the candidate pool; they cannot recommend long-tail items that never reach the pool. As the proportion of available long-tail candidates decreases, the maximum achievable APLT@10 necessarily declines, regardless of the reranking strategy employed. Put differently, a user's top-100 candidate pool contains, on average, approximately 18 long-tail items at $\alpha = 1.0$ on MovieLens-1M, but only about 5 at $\alpha = 2.0$. This available supply forms the effective *budget* from which any reranking method must construct a fairer top-10 recommendation list.

Positional Distribution of Long-Tail Items in the Pool

The pool long-tail fraction measures *how many* tail items survive in the pool, but not *where* they sit. Position matters for post-processing mitigation strategies. For example, an item ranked 95th must displace far more higher-scored candidates to enter the top-10 than an item ranked 15th, at a correspondingly higher relevance cost. To separate these two effects, we

partition the 100-item candidate pool into three zones:

- **Z-1** (positions 1–10): items already in the unmitigated top-10; recommending them requires no intervention;
- **Z-2** (positions 11–50): the *promotable* zone, from which reranking can draw at moderate relevance cost;
- **Z-3** (positions 51–100): the *deep pool*, where the base model's relevance estimates are weakest and promotion is most expensive.

The defined zones are used throughout the remainder of the chapter, in particular in the zone-migration analysis of Section 4.5 and the cross-architecture comparison of Section 4.7. For α level, Tables 4.4 and 4.5 show the mean number of long-tail items per user in each zone together with their mean rank within that zone for each . Furthermore, Figure 4.3 shows the corresponding positional density across the full pool.

Table 4.4: MovieLens: mean long-tail items per user and their mean rank within each zone of the 100-candidate pool (format: *count* (mean rank)). Z-1 = positions 1–10 (baseline top-10), Z-2 = positions 11–50 (promotable), Z-3 = positions 51–100 (deep pool).

α	Z-1 (1–10) count (mean rank)	Z-2 (11–50) count (mean rank)	Z-3 (51–100) count (mean rank)
1.00	1.89 (5.3)	6.94 (30.5)	9.01 (75.8)
1.20	1.33 (5.3)	5.10 (30.7)	6.91 (75.9)
1.50	0.85 (5.4)	3.39 (30.8)	4.78 (76.1)
1.75	0.60 (5.4)	2.40 (30.9)	3.52 (76.2)
2.00	0.43 (5.4)	1.74 (31.2)	2.62 (76.4)

Two main observations emerge from the tables. First, long-tail items are strongly reduced in every zone. On MovieLens-1M, the average number of long-tail items per user drops from 1.89 to 0.43 in Z-1 (-77%), from 6.94 to 1.74 in Z-2 (-75%), and from 9.01 to 2.62 in Z-3 (-71%) as α increases from 1.0 to 2.0. Amazon Books shows the same pattern, starting from an even lower value in Z-1 ($1.02 \rightarrow 0.24$). Notably, Z-2—arguably the most important zone for reranking methods—loses about three quarters of its long-tail items.

Second, despite this sharp drop in counts, the average rank of the remaining long-tail items within each zone stays almost unchanged. On MovieLens-1M, the mean rank in Z-2 shifts only slightly from 30.5 to 31.2, while in Z-1

Table 4.5: Amazon Books: mean long-tail items per user and their mean rank within each zone of the 100-candidate pool.

α	Z-1 (1–10) count (mean rank)	Z-2 (11–50) count (mean rank)	Z-3 (51–100) count (mean rank)
1.00	1.02 (5.8)	5.61 (31.4)	8.44 (76.1)
1.20	0.76 (5.8)	4.35 (31.5)	6.78 (76.2)
1.50	0.48 (5.9)	2.96 (31.7)	4.83 (76.3)
1.75	0.35 (5.9)	2.18 (31.9)	3.66 (76.4)
2.00	0.24 (6.0)	1.56 (32.0)	2.68 (76.5)

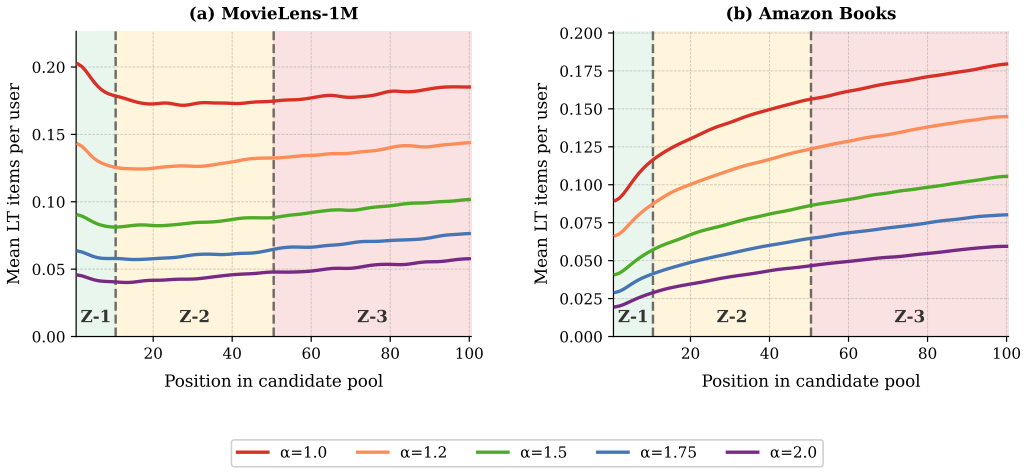


Figure 4.3: Positional density of long-tail items in the 100-item candidate pool for (a) MovieLens-1M and (b) Amazon Books, one curve per bias level α. Shaded backgrounds mark the three zones Z-1 (1–10), Z-2 (11–50), and Z-3 (51–100). Increasing α lowers the curves at every position; on Amazon Books the density additionally rises with pool depth, indicating that surviving tail items skew toward the deep pool.

it changes from 5.3 to 5.4. Amazon Books shows a similar pattern (Z-2: 31.4 $\rightarrow$ 32.0). This indicates that popularity bias does not gradually push long-tail items to worse ranks within the candidate pool. Instead, it removes many of them entirely, while the remaining items keep roughly the same rank distribution.

As a result, the main limitation is not where long-tail items appear, but how many are available. As α increases, post-processing methods have fewer candidates to promote, rather than equally many candidates at worse positions. Figure 4.3 illustrates this clearly: the curves shift downward with increasing α, instead of tilting to the right.

This distinction directly affects the mitigation results discussed next. Methods that must fill a fixed quota of long-tail items in the top-10—such as FA*IR with $\tau = 0.9$—will first use up the shrinking supply in Z-2 and then rely on Z-3, where relevance estimates are less reliable. Section 4.5 examines this shift in detail.

4.2 Base-Model Baseline Performance

Before evaluating any mitigation strategy, we characterize the behavior of the unmitigated BPR base model across bias levels. Table 4.6 reports NDCG@10 and APLT@10 on both datasets (mean $\pm$ std over three seeds).

Table 4.6: BPR baseline metrics at @10 (mean $\pm$ std over 3 seeds).

	MovieLens-1M		Amazon Books	
α	NDCG@10	APLT@10	NDCG@10	APLT@10
1.00	0.0352 ± 0.0012	0.1914 ± 0.0023	0.0060 ± 0.0001	0.1027 ± 0.0012
1.20	0.0427 ± 0.0014	0.1328 ± 0.0044	0.0075 ± 0.0001	0.0759 ± 0.0040
1.50	0.0545 ± 0.0011	0.0849 ± 0.0014	0.0102 ± 0.0002	0.0480 ± 0.0008
1.75	0.0612 ± 0.0016	0.0596 ± 0.0012	0.0105 ± 0.0001	0.0346 ± 0.0008
2.00	0.0692 ± 0.0017	0.0425 ± 0.0003	0.0096 ± 0.0004	0.0248 ± 0.0014

Accuracy (NDCG@10). On both datasets, NDCG@10 increases as α rises. For MovieLens-1M, NDCG@10 improves from 0.0352 to 0.0692, while for Amazon Books it increases from 0.0060 to 0.0105. This improvement is a direct consequence of the increasing popularity bias in the training data. As α grows, the model concentrates more of its predictive capacity on a small set of highly popular items, which also account for a large proportion of the held-out

test interactions. On Amazon Books, however, NDCG@10 decreases slightly between $\alpha = 1.75$ and $\alpha = 2.0$ ($0.0105 \rightarrow 0.0096$), suggesting that excessive concentration eventually reduces recommendation quality even within the head of the distribution. At extreme bias levels, relevant mid-popularity items are increasingly displaced alongside long-tail items, reducing the model's overall discriminative capability.

Fairness (APLT@10). In contrast, APLT@10 declines monotonically with increasing α on both datasets, falling from 0.191 to 0.043 on MovieLens-1M (-78%) and from 0.103 to 0.025 on Amazon Books (-76%). This decline follows directly from the reduction in long-tail representation within the candidate pool reported in Table 4.3. As popularity bias intensifies, the BPR model learns increasingly weaker representations for long-tail items, causing them to appear less frequently in the top-100 candidate pool. The consistently small standard deviations across random seeds indicate that this effect is stable and reproducible rather than an artifact of initialization.

The accuracy–fairness trade-off. Taken together, these results reveal a clear divergence between accuracy and fairness: as α increases, NDCG@10 improves while APLT@10 deteriorates. This pattern highlights the central challenge faced by post-processing mitigation methods. The base model becomes increasingly accurate according to conventional ranking metrics, yet simultaneously becomes less capable of exposing long-tail content. In other words, the conditions under which the recommender appears most effective are also those under which it is least fair.

This trade-off motivates the central research question of the thesis: to what extent can post-processing methods recover the long-tail exposure lost to popularity bias, and how does that recovery capability change as the outputs of the underlying recommender become increasingly biased?

4.3 Nash Parameter Selection

Each mitigation method has one trade-off parameter: λ for the two xQuAD variants and τ for FA*IR. This parameter controls how aggressively long-tail items are promoted. All the results that follow depend on these parameters being chosen in a principled, reproducible way, rather than being tuned opportunistically.

We therefore select parameters using a multi-objective criterion: for each method, dataset, and bias level, we choose the parameter value that gives the best Nash score following our methodology 3.6.

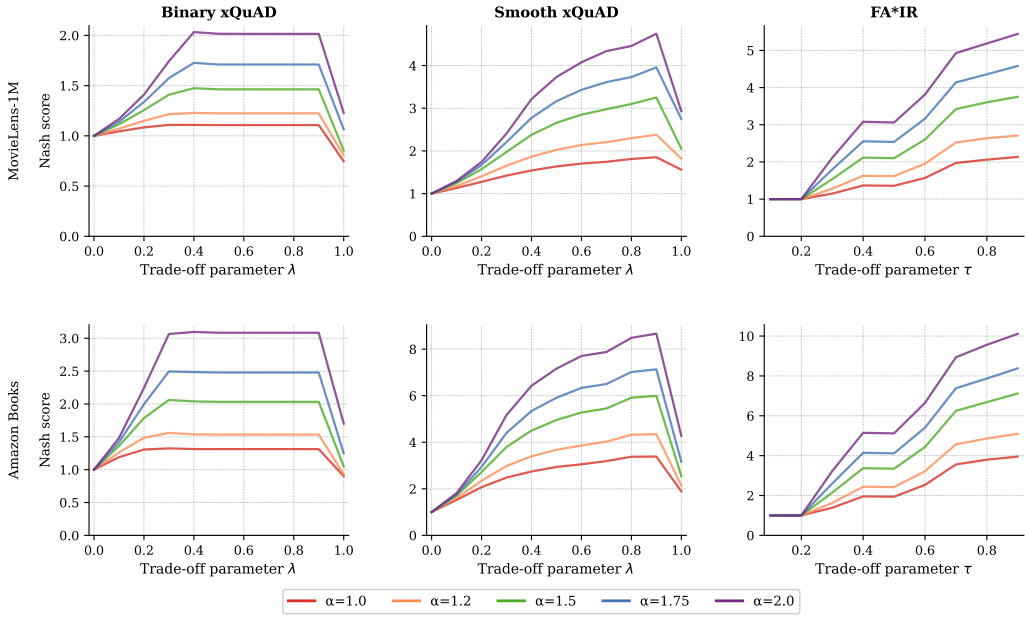


Figure 4.4: Nash score as a function of the trade-off parameter for Binary xQuAD, Smooth xQuAD, and FA*IR on MovieLens-1M (top row) and Amazon Books (bottom row), evaluated at all five bias levels α.

Figure 4.4 shows the resulting Nash curves. The parameter maximizing the mean Nash score across all five α levels is selected as the cross-α consensus; Each method exhibits a clearly identifiable optimum value and Table 4.7 reports the consensus values used in all subsequent experiments.

Table 4.7: Nash-selected parameters (cross-α consensus from the BPR sensitivity analysis). These values are reused unchanged in all cross-model experiments.

Method	MovieLens-1M	Amazon Books
Binary xQuAD (λ^*)	0.4	0.3
Smooth xQuAD (λ^*)	0.9	0.9
FA*IR (τ^*)	0.9	0.9

The chosen parameters are stable across datasets. Smooth xQuAD and FA*IR both select the upper end of their parameter ranges ($\lambda^* = 0.9$ and $\tau^* = 0.9$ on both datasets). This matches the Nash curves in Figure 4.4, which rise steadily and peak just before relevance starts to collapse at the extreme end of the range.

Binary xQuAD selects $\lambda^* = 0.4$ on MovieLens-1M and $\lambda^* = 0.3$ on Amazon Books. However, the curves show that both values lie on a broad plateau spanning roughly $\lambda \in [0.3, 0.9]$, where the ranking does not change. This plateau occurs because of the binary discount: once every long-tail item in the candidate pool has been promoted past the head items it can displace, increasing λ further has no effect. Parameter choices within this plateau are therefore equivalent, and the apparent difference between datasets is not meaningful.

Per-level stability. Tables 4.8 and 4.9 show the best parameter values for each bias level for the mitigation strategies. Smooth xQuAD and FA*IR are fully stable: they choose the same parameters at every bias level on both datasets ($\lambda^* = \tau^* = 0.9$ in all cases).

Binary xQuAD is also mostly stable, with only one small deviation on each dataset. On MovieLens-1M, it selects $\lambda^* = 0.3$ at $\alpha = 1.0$ and on Amazon Books, it selects $\lambda^* = 0.4$ at $\alpha = 2.0$. However, both cases lie in the flat region as observed in Figure 4.4.

Table 4.8: MovieLens-1M: Nash-selected parameter (λ^* or τ^*) and corresponding Nash score (in parentheses) at each bias level α. The Nash score balances accuracy and fairness improvements over the baseline. Scores rise with α because the baseline APLT collapses, inflating the relative gain (growing headroom); this does not indicate an improving absolute trade-off (see Section 4.3). Parameters are stable across α up to plateau equivalence, confirming that a single calibration suffices.

Method	$\alpha = 1.0$	$\alpha = 1.2$	$\alpha = 1.5$	$\alpha = 1.75$	$\alpha = 2.0$
Binary (λ^*)	0.3 (1.109)	0.4 (1.228)	0.4 (1.474)	0.4 (1.726)	0.4 (2.032)
Smooth (λ^*)	0.9 (1.852)	0.9 (2.379)	0.9 (3.253)	0.9 (3.956)	0.9 (4.747)
FA*IR (τ^*)	0.9 (2.136)	0.9 (2.709)	0.9 (3.752)	0.9 (4.582)	0.9 (5.444)

Interpreting the rising Nash scores. A small but important point in Tables 4.8 and 4.9 is that the best Nash scores rise strongly as α increases, for example from $1.85 \rightarrow 4.75$ for Smooth xQuAD on MovieLens-1M and from $3.95 \rightarrow 10.11$ for FA*IR on Amazon Books. At first glance, this may look like the trade-off is getting better under stronger bias. But this is not the right interpretation. The higher Nash scores do not mean that the absolute trade-off is improving. They mean that the relative gain above the baseline is increasing, because the baseline APLT becomes very small

Table 4.9: Amazon Books: Nash-selected parameter (λ^* or τ^*) and corresponding Nash score (in parentheses) at each bias level α. The Nash score balances accuracy and fairness improvements over the baseline. Scores rise with α because the baseline APLT collapses, inflating the relative gain (growing headroom); this does not indicate an improving absolute trade-off (see Section 4.3). Parameters are stable across α up to plateau equivalence, confirming that a single calibration suffices.

Method	$\alpha = 1.0$	$\alpha = 1.2$	$\alpha = 1.5$	$\alpha = 1.75$	$\alpha = 2.0$
Binary (λ^*)	0.3 (1.324)	0.3 (1.559)	0.3 (2.061)	0.3 (2.494)	0.4 (3.096)
Smooth (λ^*)	0.9 (3.385)	0.9 (4.347)	0.9 (5.995)	0.9 (7.133)	0.9 (8.664)
FA*IR (τ^*)	0.9 (3.953)	0.9 (5.098)	0.9 (7.122)	0.9 (8.380)	0.9 (10.112)

as α grows.Figure 4.5 clearly shows this trend. Since the Nash score in Equation 3.15 is based on relative improvement, a near-zero baseline can make even a modest absolute gain look very large.

So, the rising scores reflect more relative headroom, not a better absolute trade-off. In fact, the best possible APLT gets much lower as α increases, as shown in Section 4.4.

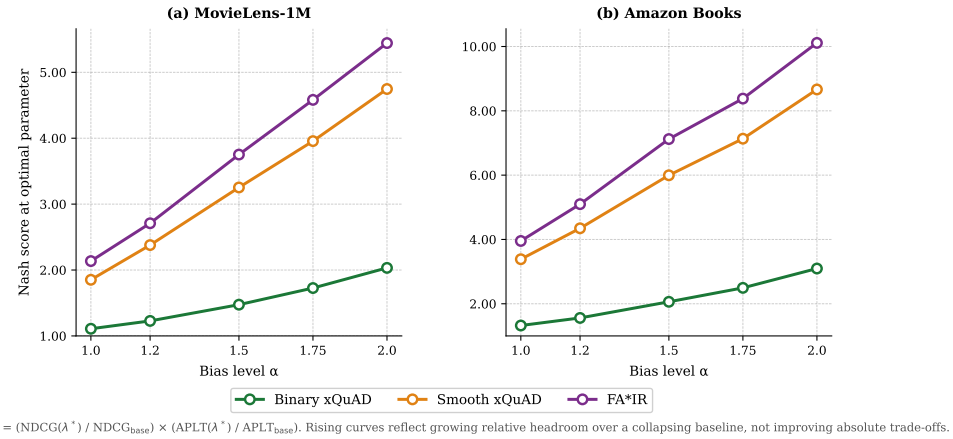


Figure 4.5: Nash score at the optimal parameter as a function of bias level α for (a) MovieLens-1M and (b) Amazon Books. The rising curves reflect growing *relative* headroom over a baseline whose APLT collapses with α (Table 4.6); they do not indicate improving absolute trade-offs, which decline as shown in Section 4.4.

4.4 Mitigation Effectiveness and the Structural Ceiling

This section presents the main result of the thesis. First, we establish the effectiveness ranking of the three mitigation methods at their Nash-selected parameters (Section 4.4). Then, we show that the maximum achievable long-tail exposure decreases with bias for *every* method, which we call the structural ceiling (Section 4.4). Finally, we illustrate the same effect in accuracy–fairness trade-off space (Section 4.4).

Mitigation Effectiveness at Nash Parameters

Tables 4.10 and 4.11 report APLT@10 and NDCG@10 at the BPR baseline and at Nash-optimal parameters for all three methods across the five α levels. The NDCG@10 rows quantify the accuracy cost incurred for each fairness gain.

Table 4.10: MovieLens-1M: APLT@10 and NDCG@10 at Nash parameters (Binary $\lambda = 0.4$, Smooth $\lambda = 0.9$, FA*IR $\tau = 0.9$). The highest APLT@10 in each row is bolded.

α	Metric	Baseline	Binary xQuAD	Smooth xQuAD	FA*IR
1.00	APLT@10	0.191	0.224	0.449	**0.605**
	NDCG@10	0.035	0.033	0.028	0.023
1.20	APLT@10	0.133	0.174	0.410	**0.540**
	NDCG@10	0.043	0.040	0.033	0.030
1.50	APLT@10	0.085	0.134	0.348	**0.457**
	NDCG@10	0.055	0.051	0.043	0.038
1.75	APLT@10	0.060	0.110	0.291	**0.390**
	NDCG@10	0.061	0.057	0.050	0.045
2.00	APLT@10	0.043	0.092	0.242	**0.290**
	NDCG@10	0.069	0.065	0.058	0.056

All three methods improve APLT@10 over the BPR baseline at every bias level and on both datasets. The effectiveness ordering is stable: FA*IR achieves the largest absolute APLT gain in all ten conditions, followed by Smooth xQuAD and then Binary xQuAD. The accuracy cost follows the same pattern but in reverse. FA*IR has the largest NDCG@10 drop and

Table 4.11: Amazon Books: APLT@10 and NDCG@10 at Nash parameters (Binary $\lambda = 0.3$, Smooth $\lambda = 0.9$, FA*IR $\tau = 0.9$). The highest APLT@10 in each row is bolded.

α	Metric	Baseline	Binary xQuAD	Smooth xQuAD	FA*IR
1.00	APLT@10	0.103	0.179	0.464	**0.628**
	NDCG@10	0.006	0.006	0.004	0.004
1.20	APLT@10	0.076	0.127	0.429	**0.560**
	NDCG@10	0.008	0.007	0.006	0.005
1.50	APLT@10	0.048	0.106	0.357	**0.457**
	NDCG@10	0.010	0.010	0.008	0.007
1.75	APLT@10	0.035	0.093	0.301	**0.382**
	NDCG@10	0.011	0.010	0.009	0.008
2.00	APLT@10	0.025	0.089	0.245	**0.298**
	NDCG@10	0.010	0.009	0.008	0.007

Binary xQuAD the smallest. This reflect the progressively stricter exposure constraint each method imposes.

At $\alpha = 1.0$ on MovieLens-1M, for example, FA*IR increases APLT from 0.191 to 0.605 but lowers NDCG from 0.035 to 0.023. In contrast, Binary xQuAD makes only a small fairness improvement (0.191 $\rightarrow$ 0.224) while keeping almost all of the accuracy. The three methods therefore occupy different, consistently ordered positions on the accuracy–fairness spectrum.

The Structural Ceiling

The central finding of this study is that the maximum achievable APLT@10 declines monotonically with α for *all* three methods and the baseline alike. Figure 4.6 shows this for BPR.

The defining feature of the figure is that all four curves descend in parallel: increasing bias lowers not only the baseline's long-tail exposure but the ceiling reachable by every mitigation method. The relative declines from $\alpha = 1.0$ to $\alpha = 2.0$ are substantial and consistent across datasets:

- MovieLens-1M: FA*IR 0.605 $\rightarrow$ 0.290 (-52%); baseline 0.191 $\rightarrow$ 0.043 (-78%).
- Amazon Books: FA*IR 0.628 $\rightarrow$ 0.298 (-54%); baseline 0.103 $\rightarrow$ 0.025 (-77%).

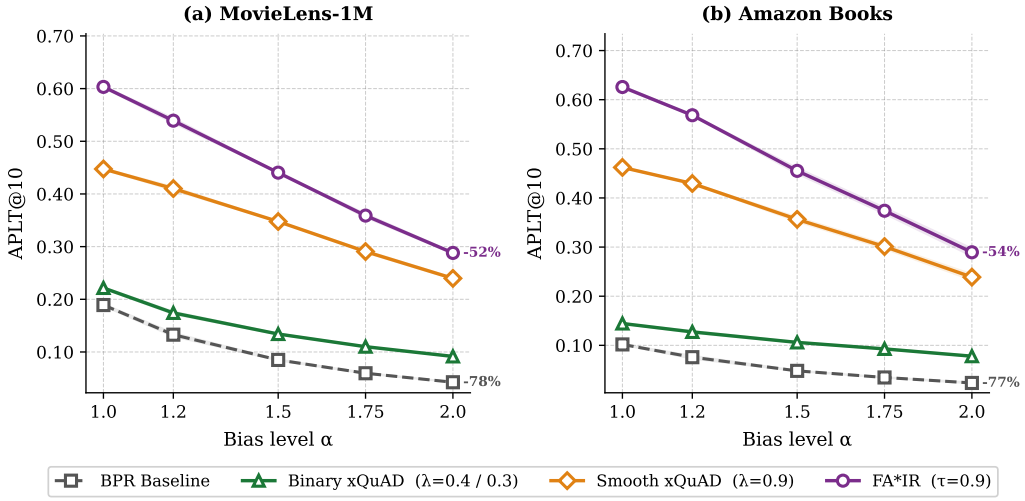


Figure 4.6: APLT@10 for BPR across all α levels at Nash-selected parameters on (a) MovieLens-1M and (b) Amazon Books. All four curves (Baseline, Binary xQuAD, Smooth xQuAD, FA*IR) decline together as α increases. Annotations at $\alpha = 2.0$ give the relative APLT decline from $\alpha = 1.0$ for FA*IR and the baseline.

Two important features of this decline are worth highlighting. First, the gap between the baseline and the best method *narrows* as bias increases: FA*IR's advantage over the baseline on MovieLens-1M drops from +0.413 APLT units at $\alpha = 1.0$ to +0.248 at $\alpha = 2.0$. This means that the best mitigation method recovers less in absolute terms exactly when the base model is most biased and mitigation is most needed.

Second, the methods themselves converge: FA*IR's lead over Smooth xQuAD on MovieLens-1M shrinks from 0.156 APLT units at $\alpha = 1.0$ to 0.048 at $\alpha = 2.0$. As the deep pool runs out, the most aggressive method has less unique room to work with and moves closer to the next-best method.

This behavior follows directly from the candidate-pool depletion shown in Section 4.1. Post-processing can only promote items that appear in the candidate pool, so the pool long-tail fraction (Table 4.3) is a hard upper bound on what any re-ranker can place in the top-10. The effect is strongest for FA*IR: with $\tau = 0.9$, it targets a long-tail proportion of 7 out of 10 top positions, but at $\alpha = 2.0$, the average user's top-100 pool contains only about 5 tail items in total. The target then becomes infeasible for many users, FA*IR uses up the available tail items, and its output and APLT moves closer to that of Smooth xQuAD.

The declining ceiling is therefore not an algorithmic weakness but a

structural result of the shrinking pool: the constraint that limits fairness lies before the re-ranker, in the biased candidate set it receives.

The Accuracy–Fairness Trade-off Under Bias

The structural ceiling can equivalently be read in accuracy–fairness trade-off space. Figure 4.7 plots, for BPR on both datasets, one frontier curve per α level, with each point on a curve corresponding to one method.

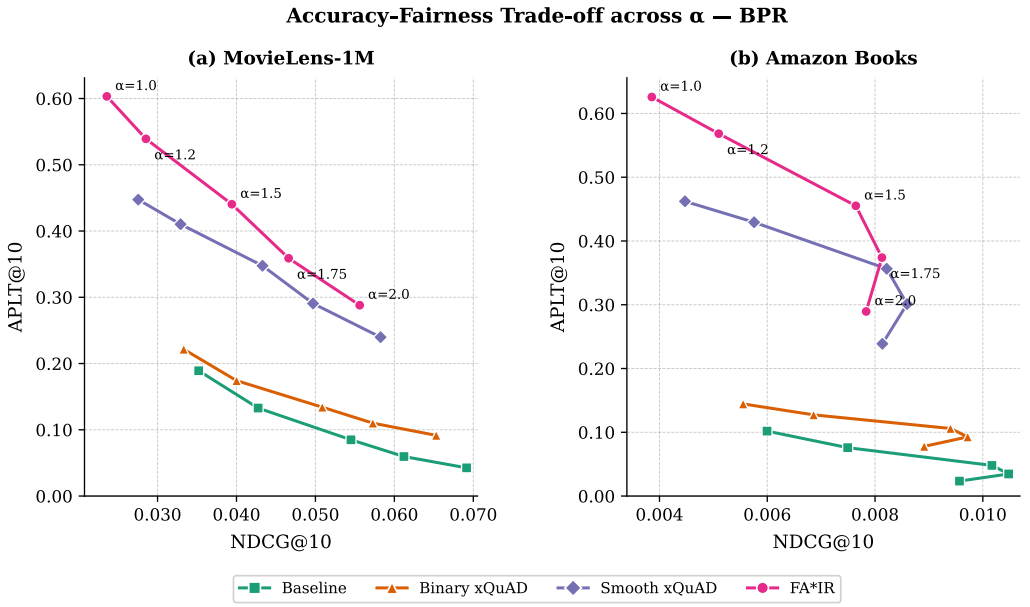


Figure 4.7: Accuracy–fairness frontier for BPR on (a) MovieLens-1M and (b) Amazon Books. Each colored curve is one α level (red = $\alpha = 1.0$ through purple = $\alpha = 2.0$); each point on a curve is one method (square = Baseline, triangle = Binary xQuAD, diamond = Smooth xQuAD, circle = FA*IR). Method names are labelled on the $\alpha = 1.0$ curve and α values at each baseline point.

Each frontier extends from FA*IR (top-left: highest APLT, lowest NDCG) to the baseline ranking (bottom-right: highest NDCG, lowest APLT), illustrating the fairness–accuracy trade-off achievable at different reranking levels. As α increases, the entire frontier shifts rightward because the baseline NDCG improves, as shown in Section 4.2, while simultaneously moving downward as the maximum attainable APLT decreases.

The most important characteristic is the *vertical span* of each frontier. Specifically, the distance between the baseline and FA*IR endpoints. This

span decreases consistently as α increases. At $\alpha = 1.0$, the four methods occupy a broad range of APLT values, meaning that the choice of mitigation method has a substantial impact on fairness outcomes. By contrast, at $\alpha = 2.0$, the same methods are clustered within a much narrower APLT range. Consequently, the influence of the mitigation method diminishes as the structural fairness ceiling becomes more restrictive. When bias is most severe, even the strongest reranking approach can achieve only limited improvements over the others. This compression of the trade-off space reflects the same structural constraint observed in the parallel decline shown in Figure 4.6, and the pattern reappears across different model architectures in Section 4.7.

Table 4.12: Nash score (Eq. 3.15), $(\text{NDCG}/\text{NDCG}_{\text{base}}) \times (\text{APLT}/\text{APLT}_{\text{base}})$, at Nash-selected parameters for BPR. The best method per column is bolded. Values rise with α because they are measured relative to a collapsing baseline (see Section 4.3); compare methods down each column, not across rows.

Dataset	Method	$\alpha = 1.0$	$\alpha = 1.2$	$\alpha = 1.5$	$\alpha = 1.75$	$\alpha = 2.0$
MovieLens-1M	FA*IR	**2.14**	**2.71**	**3.75**	**4.58**	**5.44**
	Smooth	1.85	2.38	3.25	3.96	4.75
	Binary	1.11	1.23	1.47	1.73	2.03
Amazon Books	FA*IR	**3.95**	**5.10**	**7.12**	**8.38**	**10.11**
	Smooth	3.39	4.35	6.00	7.13	8.66
	Binary	1.32	1.56	2.06	2.49	3.07

To provide a quantitative view of the frontier, Table 4.12 summarizes each method's two-dimensional operating point using the Nash score (Eq. 3.15) at every bias level. Examining the methods within each column (i.e., at a fixed α) confirms the effectiveness ranking established in Section 4.4. Across both datasets, FA*IR consistently achieves the highest balanced score, followed by Smooth xQuAD and then Binary xQuAD, while the baseline remains fixed at 1.0 by definition. The separation between methods observed in Figure 4.7 is therefore reflected in the corresponding spread of Nash scores.

The trend across rows should be interpreted in light of the discussion in Section 4.3. Although the Nash scores increase as α rises (for example, FA*IR on MovieLens-1M increases from 2.08 to 5.47), this does not indicate an improving in fairness accuracy trade-off. Instead, the increase is driven by the shrinking baseline APLT in the denominator of the Nash score. As the baseline APLT approaches zero, relative gains become larger even though the maximum achievable APLT is simultaneously declining, as shown in Figure 4.6. Consequently, Table 4.12 should be interpreted as measuring

relative separation between methods at a given bias level, whereas Figure 4.6 captures the *absolute* decline in the attainable fairness ceiling. The relationship between these two perspectives is examined in detail in Section 4.6.

4.5 Mechanism: Long-tail Zone Migration

Section 4.4 demonstrated that the APLT ceiling declines as bias increases across all mitigation methods. This section investigates the underlying cause by examining where the long-tail items appearing in the final top-10 recommendations originate within the candidate pool. As introduced in Section 4.1, the candidate pool is divided into three regions: Z-1 (positions 1–10), containing items already present in the unmitigated top-10; Z-2 (positions 11–50), a moderately accessible region from which items can be promoted at a reasonable relevance cost; and Z-3 (positions 51–100), the deep-pool region where relevance estimates are less reliable and promotion is most costly. By tracking the origin of promoted long-tail items across these zones, we can directly observe how mitigation strategies adapt as the available supply of long-tail candidates becomes increasingly constrained.

Tables 4.13 and 4.14 report the average number of long-tail items per user that appear in the final top-10 recommendations, separated according to their zone of origin for each method and bias level. Figure 4.8 presents the same information visually using stacked bar charts, where each Coloured segment represents the contribution of a particular zone to the final long-tail exposure.

The baseline rows serve as a reference point. Because the baseline applies no reranking, all long-tail items that appear in the final top-10 originate from Z-1, and their number decreases in line with the shrinking long-tail candidate pool discussed in Section 4.1. In contrast, all three mitigation methods exhibit a systematic shift toward deeper regions of the candidate pool as bias increases. The extent of this shift varies by method, revealing different levels of dependence on lower-ranked long-tail items.

Binary xQuAD: limited promotion with increasing reliance on the deep pool. Binary xQuAD makes only minor adjustments to the original ranking. Its Z-1 contribution remains almost identical to the baseline across all bias levels (MovieLens-1M: 1.887 vs. 1.891 at $\alpha = 1.0$, and 0.425 vs. 0.425 at $\alpha = 2.0$), indicating that the method introduces only a small number of additional long-tail items rather than substantially restructuring the top-10 recommendations. Nevertheless, a clear migration pattern emerges. Among the relatively few items that Binary xQuAD promotes, the proportion originating from the deep pool (Z-3) increases markedly as bias grows, rising

Table 4.13: MovieLens: mean long-tail items per user in the final top-10, broken down by their origin zone in the 100-candidate pool. Z-1 (1–10): items already in the unmitigated top-10. Z-2 (11–50): items promoted from the reachable zone. Z-3 (51–100): items pulled from the deep pool. Reranking applied using the same implementation as the published cross-model experiments.

α	Method	Z-1 (1–10)	Z-2 (11–50)	Z-3 (51–100)	Total
1.00	Baseline	1.891	0.000	0.000	**1.891**
	Binary xQuAD	1.887	0.265	0.064	**2.216**
	Smooth xQuAD	1.715	2.031	0.728	**4.474**
	FA*IR	1.891	2.936	1.205	**6.032**
1.20	Baseline	1.328	0.000	0.000	**1.328**
	Binary xQuAD	1.326	0.317	0.099	**1.742**
	Smooth xQuAD	1.233	1.936	0.933	**4.102**
	FA*IR	1.328	2.616	1.447	**5.391**
1.50	Baseline	0.849	0.000	0.000	**0.849**
	Binary xQuAD	0.848	0.344	0.149	**1.341**
	Smooth xQuAD	0.800	1.616	1.061	**3.476**
	FA*IR	0.849	2.053	1.503	**4.405**
1.75	Baseline	0.597	0.000	0.000	**0.597**
	Binary xQuAD	0.596	0.325	0.179	**1.101**
	Smooth xQuAD	0.569	1.304	1.034	**2.907**
	FA*IR	0.597	1.590	1.404	**3.590**
2.00	Baseline	0.425	0.000	0.000	**0.425**
	Binary xQuAD	0.425	0.302	0.189	**0.916**
	Smooth xQuAD	0.409	1.028	0.961	**2.398**
	FA*IR	0.425	1.224	1.231	**2.881**

Table 4.14: Amazon Books: mean long-tail items per user in the final top-10, broken down by their origin zone in the 100-candidate pool. Z-1 (1–10): items already in the unmitigated top-10. Z-2 (11–50): items promoted from the reachable zone. Z-3 (51–100): items pulled from the deep pool. Reranking applied using the same implementation as the published cross-model experiments.

α	Method	Z-1 (1–10)	Z-2 (11–50)	Z-3 (51–100)	Total
1.00	Baseline	1.019	0.000	0.000	**1.019**
	Binary xQuAD	1.019	0.374	0.051	**1.444**
	Smooth xQuAD	1.014	2.856	0.752	**4.622**
	FA*IR	1.019	3.845	1.394	**6.258**
1.20	Baseline	0.759	0.000	0.000	**0.759**
	Binary xQuAD	0.759	0.424	0.088	**1.271**
	Smooth xQuAD	0.756	2.517	1.020	**4.294**
	FA*IR	0.759	3.220	1.704	**5.683**
1.50	Baseline	0.481	0.000	0.000	**0.481**
	Binary xQuAD	0.481	0.424	0.145	**1.051**
	Smooth xQuAD	0.479	1.911	1.175	**3.566**
	FA*IR	0.481	2.334	1.738	**4.553**
1.75	Baseline	0.347	0.000	0.000	**0.347**
	Binary xQuAD	0.347	0.395	0.171	**0.912**
	Smooth xQuAD	0.346	1.511	1.156	**3.013**
	FA*IR	0.347	1.793	1.603	**3.744**
2.00	Baseline	0.236	0.000	0.000	**0.236**
	Binary xQuAD	0.236	0.341	0.179	**0.757**
	Smooth xQuAD	0.236	1.143	1.016	**2.395**
	FA*IR	0.236	1.327	1.342	**2.905**

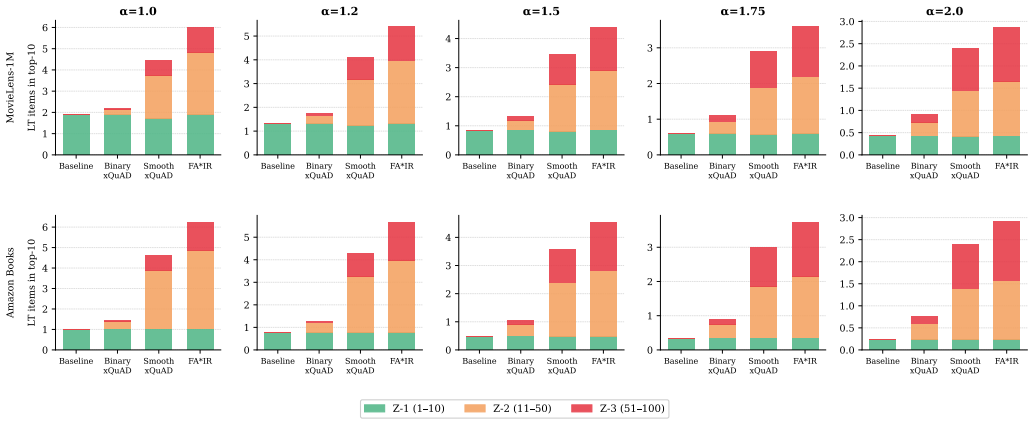


Figure 4.8: Origin-zone composition of long-tail items in the final top-10, for each method at each bias level α, on MovieLens-1M (top row) and Amazon Books (bottom row). Each bar is segmented by origin zone: Z-1 (1–10, green), Z-2 (11–50, orange), and Z-3 (51–100, red). As α increases, the red Z-3 segment grows as a share of every method's promoted mass, indicating that fairness gains are increasingly sourced from the deep pool.

from 19.5% to 38.5% on MovieLens-1M and from 12.0% to 34.4% on Amazon Books. As the supply of accessible long-tail items in Z-2 diminishes, even a conservative reranking strategy must increasingly draw candidates from deeper regions of the pool.

Smooth xQuAD: approaching a balance between Z-2 and Z-3. Smooth xQuAD performs substantially more reranking and therefore exhibits a stronger migration pattern. At low bias levels, most promoted long-tail items originate from the reachable Z-2 region (MovieLens-1M, $\alpha = 1.0$: Z-2 = 2.031 and Z-3 = 0.728). However, the difference between the two zones decreases steadily as bias increases. The Z-2 advantage narrows from 1.30 items at $\alpha = 1.0$ to just 0.07 items at $\alpha = 2.0$, where Z-2 contributes 1.028 items and Z-3 contributes 0.961. Correspondingly, the Z-3-to-Z-2 ratio increases from 0.36 to 0.94 across the bias range. The same behavior is observed on Amazon Books, where the Z-2 margin contracts from 2.10 to 0.13. Although Smooth xQuAD does not fully transition to relying on Z-3, it approaches a point where the reachable and deep-pool regions contribute almost equally.

FA*IR: transition to deep-pool dominance. FA*IR exhibits the strongest migration toward deeper pool regions and is the only method to complete the transition. On MovieLens-1M, the difference between Z-2 and Z-3 contribu-

tions decreases monotonically from +1.73 at $\alpha = 1.0$ to −0.01 at $\alpha = 2.0$. At the highest bias level, FA*IR draws slightly more promoted long-tail items from Z-3 than from Z-2 (Z-3 = 1.231 vs. Z-2 = 1.224). An identical crossover occurs on Amazon Books, where Z-3 contributes 1.342 items compared with 1.327 from Z-2. Expressed as a proportion of all promoted long-tail items, the contribution of Z-3 increases from approximately one-quarter at $\alpha = 1.0$ (29.1% on MovieLens-1M and 26.6% on Amazon Books) to roughly one-half at $\alpha = 2.0$ (50.1% and 50.3%, respectively). This transition is visible in Figure 4.8 as the expansion of the red Z-3 segment. The fact that the crossover occurs at the same bias level on two datasets with very different characteristics suggests that it reflects a structural property of fairness reranking under candidate-pool depletion rather than a dataset-specific effect.

Interpretation. Together, the three methods reveal a common underlying mechanism operating at different levels of intensity. As increasing bias reduces the supply of reachable long-tail items in Z-2, maintaining fairness objectives requires methods to promote candidates from progressively deeper regions of the pool. Because these Z-3 items are ranked much lower by the base recommender, they are associated with weaker relevance estimates and higher accuracy costs. Consequently, the structural ceiling described in Section 4.4 is not only lower under stronger bias but is also composed of increasingly lower-ranked long-tail items. The fairness that remains attainable therefore becomes progressively more expensive to achieve. Mitigation methods are not paying a fixed accuracy cost for a fixed fairness gain; instead, the cost escalates as they are forced deeper into the candidate pool. The next section quantifies this escalation through efficiency, relative improvement, and marginal sensitivity analyses (Section 4.6), while Section 4.7 demonstrates that the architectures experiencing the most rapid pool depletion are also those with the steepest declines in their attainable fairness ceilings.

4.6 Quantifying the Trade-off

Sections 4.4 and 4.5 have shown that achievable long-tail exposure decreases as bias increases, and that the remaining exposure is increasingly drawn from deeper, lower-relevance regions of the candidate pool. This section translates these qualitative observations into three quantitative, deployment-relevant metrics: the efficiency of each method as an exchange rate between accuracy and fairness (Section 4.6); the relative improvement over the baseline, which we demonstrate becomes increasingly misleading under high bias (Section 4.6); and the marginal cost associated with each incremental increase in bias

(Section 4.6).

Efficiency: the Accuracy–Fairness Exchange Rate

We define mitigation efficiency as the APLT@10 gain obtained per 1% of NDCG@10 sacrificed relative to the baseline. This acts as an exchange rate that quantifies how effectively a method converts predictive accuracy into long-tail exposure. Higher values therefore indicate more favorable trade-offs between fairness and accuracy.

Table 4.15 reports efficiency values averaged across the three architectures, while the per-architecture breakdown is provided in Appendix A.1 (Tables A.1 and A.2). Figure 4.9 visualizes these averaged results.

Table 4.15: Mitigation efficiency (APLT gain per 1% NDCG cost) averaged across BPR, NeuMF, and LightGCN. Higher values indicate a more favourable accuracy–fairness trade-off. Smooth xQuAD and FA*IR are statistically indistinguishable in efficiency at every bias level (differences ≤ 0.0001), indicating a shared accuracy–fairness exchange rate; Binary xQuAD is less efficient at low bias but its efficiency rises with α.

α	Method	MovieLens-1M	Amazon Books
1.00	Binary xQuAD	0.0074	0.0095
	Smooth xQuAD	0.0136	0.0159
	FA*IR	0.0137	0.0160
1.20	Binary xQuAD	0.0077	0.0104
	Smooth xQuAD	0.0134	0.0170
	FA*IR	0.0134	0.0171
1.50	Binary xQuAD	0.0084	0.0136
	Smooth xQuAD	0.0134	0.0170
	FA*IR	0.0134	0.0170
1.75	Binary xQuAD	0.0091	0.0139
	Smooth xQuAD	0.0132	0.0157
	FA*IR	0.0133	0.0158
2.00	Binary xQuAD	0.0104	0.0135
	Smooth xQuAD	0.0132	0.0147
	FA*IR	0.0131	0.0147

The central observation is that Smooth xQuAD and FA*IR achieve essentially identical fairness efficiency, indicating that they operate on the

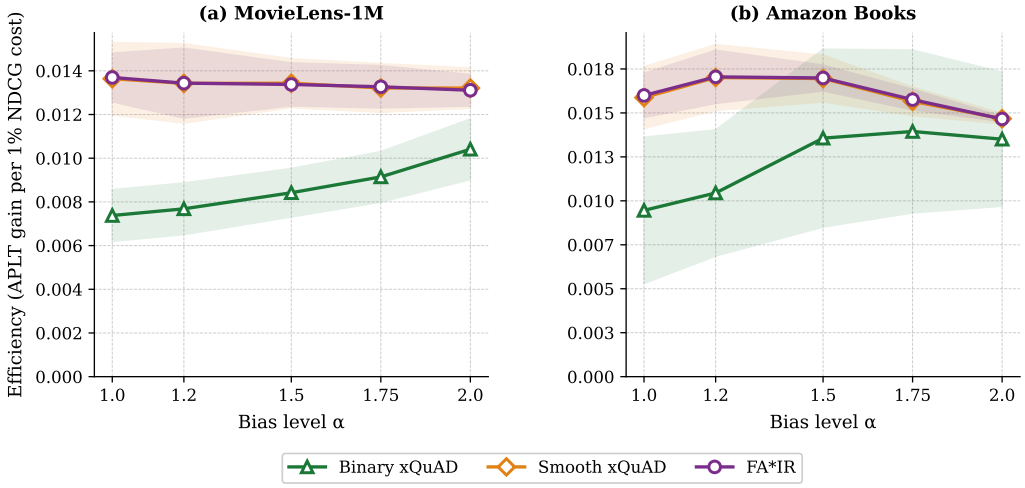


Figure 4.9: Mitigation efficiency (APLT gain per 1% NDCG cost) averaged across BPR, NeuMF, and LightGCN, for (a) MovieLens-1M and (b) Amazon Books. Shaded bands show ±1 std across architectures. Smooth xQuAD and FA*IR are indistinguishable at every bias level; Binary xQuAD is less efficient but its efficiency rises with α.

same underlying exchange rate between accuracy loss and long-tail exposure gain. Their efficiency values match to within 0.0001 across all bias levels and both datasets (MovieLens-1M, $\alpha = 1.0$: 0.0136 vs. 0.0137; Amazon Books, $\alpha = 1.0$: 0.0159 vs. 0.0160), and this near-perfect alignment is reflected in the overlapping curves in Figure 4.9. While FA*IR consistently achieves higher absolute APLT than Smooth xQuAD (Section 4.4), it does not do so at a different efficiency rate. In practical terms, the two methods lie on the same trade-off frontier, differing only in how far they move along it rather than in how efficiently they traverse it.

Binary xQuAD exhibits a distinct behavior. It is substantially less efficient at low bias (MovieLens-1M, $\alpha = 1.0$: 0.0074, approximately half the efficiency of the other methods), but its efficiency increases with bias, reaching 0.0104 at $\alpha = 2.0$ on MovieLens-1M. This upward trend is consistent with the zone migration analysis in Section 4.5: because Binary xQuAD promotes only a small number of items and rarely draws from the deep pool, it avoids the escalating relevance penalty associated with Z-3 sourcing. As bias increases and the pool becomes more constrained, this limited exposure strategy results in a relatively smaller incremental accuracy cost per unit of fairness gained.

A normalization caveat. The architecture-level results in Appendix A.1 indicate that LightGCN achieves the highest nominal efficiency across most bias levels. However, this does not imply superior mitigation behavior. Since efficiency is defined using NDCG loss normalized by the baseline, architectures with higher baseline NDCG (such as LightGCN) exhibit smaller percentage losses for comparable absolute drops, mechanically inflating the efficiency ratio. This normalization effect is disentangled in Section 4.7, where absolute APLT gains provide a scale-invariant comparison and lead to a different ranking of architectures.

The Relative-Improvement Paradox

Fairness gains are frequently reported in the literature as relative improvements over a baseline. Table 4.16 shows why this convention becomes actively misleading in the high-bias regime.

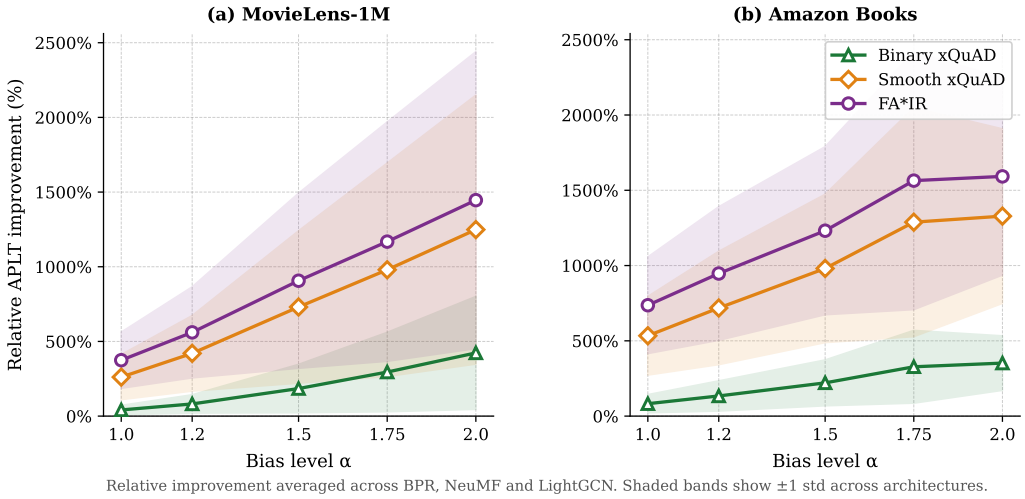


Figure 4.10: Relative APLT@10 improvement over the baseline (%), averaged across architectures, for (a) MovieLens-1M and (b) Amazon Books. Shaded bands show ±1 std across architectures. All methods' relative gains rise steeply with α even though their absolute gains shrink, because the baseline denominator collapses faster than the numerator.

Read naively, the table suggests that mitigation becomes dramatically *more* effective as bias increases. For example, Binary xQuAD's relative APLT improvement on MovieLens-1M rises from 41.3% at $\alpha = 1.0$ to 423.1% at $\alpha = 2.0$, and all methods exhibit monotonically increasing relative gains across bias levels on both datasets. This pattern is inconsistent with the

Table 4.16: Relative APLT improvement (%) at Nash-selected parameters, averaged across BPR, NeuMF, and LightGCN. Binary xQuAD shows a paradoxical increase in relative gain as α rises: the baseline APLT collapses faster than Binary xQuAD's modest absolute gain, inflating the relative figure. The same inflation affects all three methods: every method's relative gain rises with α even though absolute APLT ceilings fall (Section 4.6), making relative improvement an unreliable figure of merit under popularity bias.

α	Method	MovieLens-1M (%)	Amazon Books (%)
1.00	Binary xQuAD	41.3%	82.4%
	Smooth xQuAD	261.1%	533.8%
	FA*IR	374.4%	736.4%
1.20	Binary xQuAD	81.5%	133.9%
	Smooth xQuAD	420.2%	718.2%
	FA*IR	560.7%	947.1%
1.50	Binary xQuAD	184.4%	220.8%
	Smooth xQuAD	730.5%	980.7%
	FA*IR	906.5%	1231.5%
1.75	Binary xQuAD	294.9%	327.6%
	Smooth xQuAD	979.7%	1288.2%
	FA*IR	1168.2%	1564.1%
2.00	Binary xQuAD	423.1%	352.1%
	Smooth xQuAD	1248.7%	1328.3%
	FA*IR	1445.9%	1592.2%

structural ceiling behavior established throughout this chapter and is, in fact, an artifact of the evaluation metric.

The explanation lies in the collapsing denominator. Binary xQuAD's *absolute* APLT gain remains nearly constant across bias levels. For example, on MovieLens-1M it changes only from 0.033 ($0.191 \rightarrow 0.224$) at $\alpha = 1.0$ to 0.049 ($0.043 \rightarrow 0.092$) at $\alpha = 2.0$. The baseline APLT against which it is measured drops sharply from 0.191 to 0.043. As a result, dividing a roughly stable numerator by a shrinking denominator produces the appearance of a tenfold increase in improvement, even though the underlying absolute gain is small and only modestly changing. The same effect inflates the reported relative gains for Smooth xQuAD and FA*IR, whose percentages exceed 1,400% at $\alpha = 2.0$ despite their absolute APLT ceilings having approximately halved.

This phenomenon is a direct instance of the denominator effect discussed in Section 4.3, here expressed in its most extreme form. It carries an important methodological implication: *relative* long-tail improvement is not a reliable indicator under strong popularity bias. Methods may exhibit rapidly increasing percentage gains while their absolute contributions remain flat or even deteriorate, and approaches with substantially different absolute performance can appear deceptively similar under a relative metric. For this reason, evaluations of fairness-aware recommendation under biased conditions should priorities absolute APLT, or report it alongside relative measures. In this work, absolute APLT is used as the primary metric precisely to avoid this distortion.

Marginal Sensitivity to Bias

The final measure examines how the collapse of the APLT ceiling is distributed across the bias range: whether each increment in α imposes a uniform cost, or whether certain transitions are disproportionately damaging. Table 4.17 and Figure 4.11 report the marginal decline in APLT at each bias step, averaged across architectures.

Two patterns emerge. First, the per-step APLT loss is strictly ordered by method aggressiveness at every bias transition: FA*IR exhibits the largest decline per increment in α, followed by Smooth xQuAD, then Binary xQuAD, while the baseline shows the smallest absolute changes. On MovieLens-1M, FA*IR reaches a maximum single-step loss of 0.110 APLT units (at the $1.2 \rightarrow 1.5$ transition), compared to 0.032 for Binary xQuAD at the same step. This ordering is consistent with the zone-migration mechanism: methods that achieve higher long-tail exposure also depend more heavily on deeper pool regions, and therefore experience larger losses when those regions become

Table 4.17: Marginal APLT decline per α step at Nash-selected parameters, averaged across BPR, NeuMF, and LightGCN. Negative values indicate a fall in the APLT ceiling between consecutive bias levels; the largest decline in each row is bolded. FA*IR loses the most at every step on both datasets. The method declines peak at the 1.2 → 1.5 step — where the candidate-pool long-tail fraction drops most sharply (Table 4.3) — and decelerate thereafter, while the baseline's decline is concentrated in the earliest steps and flattens as its APLT approaches zero.

Step	Dataset	Baseline	Binary	Smooth	FA*IR
1.0 → 1.2	MovieLens-1M	−0.0473	−0.0390	−0.0527	**−0.0843**
	Amazon Books	−0.0217	−0.0161	−0.0469	**−0.0756**
1.2 → 1.5	MovieLens-1M	−0.0364	−0.0323	−0.0748	**−0.1104**
	Amazon Books	−0.0218	−0.0226	−0.0838	**−0.1224**
1.5 → 1.75	MovieLens-1M	−0.0169	−0.0203	−0.0653	**−0.0883**
	Amazon Books	−0.0111	−0.0158	−0.0623	**−0.0856**
1.75 → 2.0	MovieLens-1M	−0.0104	−0.0162	−0.0496	**−0.0648**
	Amazon Books	−0.0069	−0.0146	−0.0549	**−0.0715**

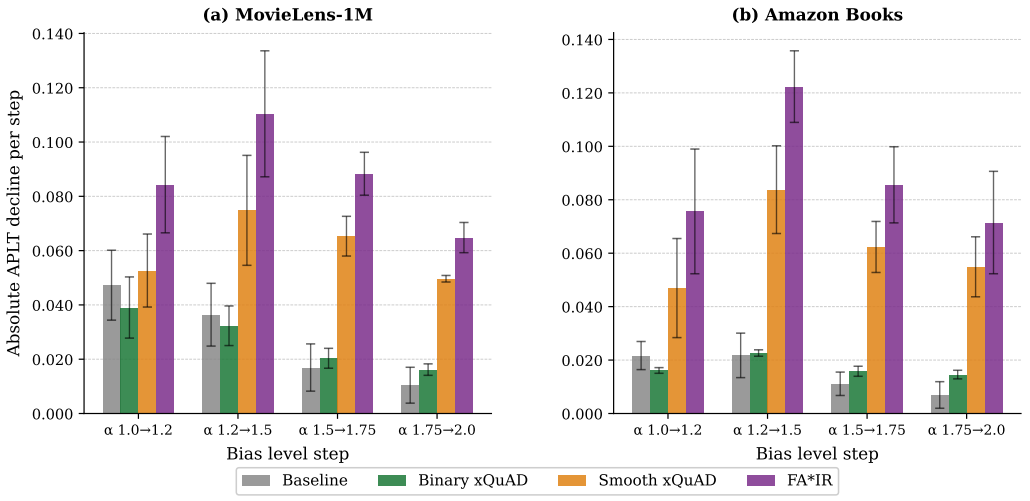


Figure 4.11: Absolute APLT decline at each bias step, averaged across BPR, NeuMF, and LightGCN, for (a) MovieLens-1M and (b) Amazon Books. Error bars show ±1 std across architectures. Larger bars indicate a faster ceiling collapse at that step. FA*IR loses the most at every step, and the method declines peak at the 1.2 → 1.5 step.

unavailable as bias increases.

Second, the decline pattern is non-linear in α and exhibits a pronounced peak at the $1.2 \rightarrow 1.5$ transition, followed by a gradual deceleration thereafter (FA*IR on MovieLens-1M: $0.084 \rightarrow 0.110 \rightarrow 0.088 \rightarrow 0.065$; Amazon Books shows a sharper peak: $0.076 \rightarrow 0.122 \rightarrow 0.086 \rightarrow 0.072$). This concave profile aligns with the candidate-pool dynamics described in Section 4.1, where the largest absolute reductions in long-tail availability occur in the early-to-mid bias range as the Z-2 region becomes depleted (e.g., MovieLens-1M long-tail fraction drops from 13.3% to 9.0% over the $1.2 \rightarrow 1.5$ step). Once this reachable supply is largely exhausted, further increases in bias remove relatively little of the remaining usable pool, and marginal losses diminish accordingly. The baseline shows a complementary pattern, with sharper early declines that flatten as APLT approaches zero, reflecting saturation rather than structural depletion. Overall, the structural ceiling does not decrease uniformly with bias; instead, its steepest descent coincides with the depletion of the reachable pool, matching the reranking-side transition captured by the zone-migration crossover in Section 4.5.

4.7 Cross-Model Validation

All results so far have been obtained using BPR as the base model. This section demonstrates that the structural ceiling generalizes to two additional architectures which are NeuMF and LightGCN. Further more, the results shows that the underlying architecture itself determines the severity of the ceiling collapse. The Nash-selected parameters from Section 4.3 are used without modification across all models; no architecture-specific re-tuning is performed.

Pool Depletion by Architecture

The three architectures differ in their baseline accuracy fairness positioning prior to any mitigation. BPR exhibits the lowest baseline NDCG but the highest baseline APLT, indicating a tendency to recommend relatively more long-tail items even without explicit intervention. In contrast, LightGCN occupies the opposite extreme highest NDCG and lowest APLT. This can be attributed to its architecture. LightGCN uses a graph-based architecture that amplifies popular-item representations through multi-hop neighborhood aggregation, thereby concentrating recommendations in the head of the distribution. NeuMF lies between these two extremes. This accuracy fairness inversion at the architecture level where higher predictive accuracy coincides

with lower long-tail exposure. This is consistent across both datasets and all bias settings.

To account for the resulting differences in ceiling behavior, we move beyond top-10 outcomes and examine the full candidate pool for each architecture, extending the positional analysis introduced in Section 4.1. Figure 4.12 reports the relative share of long-tail items at each pool position (normalized by the pool mean) across the five bias levels, separately for each architecture.

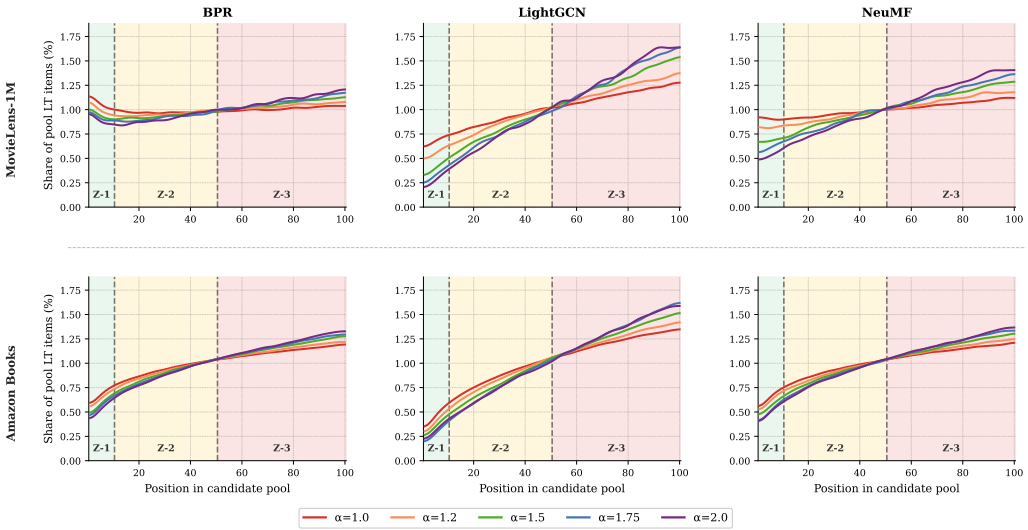


Figure 4.12: Normalized positional density of long-tail items in the candidate pool for BPR, LightGCN, and NeuMF on MovieLens-1M (top row) and Amazon Books (bottom row), one curve per bias level. Each curve is normalized to its pool mean; values below 1 in the early zones indicate that long-tail items are scarce near the top of the pool and concentrated in the deep pool. LightGCN shows the steepest rise from Z-1 to Z-3, indicating the most front-depleted pool.

The architectures differ substantially in *where* the remaining long-tail items are located within the candidate pool. For BPR, the normalized density is approximately uniform across positions: although long-tail items become increasingly scarce under higher bias, their distribution across the pool remains relatively even. LightGCN exhibits the opposite pattern, with a pronounced shift from suppressed Z-1 and Z-2 regions toward Z-3, indicating that its long-tail items are disproportionately concentrated in deep-pool positions where promotion is most costly. NeuMF again occupies an intermediate position between these two extremes.

In this sense, LightGCN's candidate pool is not only reduced in long-tail content but also *front-depleted*: the accessible Z-2 region, from which

mitigation methods primarily draw under low cost, is the most impoverished. This structural property directly explains the ceiling ordering observed in the next section, turning the empirical result that "LightGCN's ceiling declines most sharply" into a consequence of how long-tail items are distributed within its pool.

The Structural Ceiling Across Architectures

Table 4.18 reports the Smooth xQuAD APLT ceiling at the extremes of the bias range for each architecture and dataset.

Table 4.18: Smooth xQuAD APLT ceiling at $\alpha = 1.0$ and $\alpha = 2.0$ across architectures and datasets (mean over 3 seeds), with the relative decline. The decline deepens with base-model sophistication on both datasets.

Dataset	Model	APLT ($\alpha = 1.0$)	APLT ($\alpha = 2.0$)	Decline
MovieLens-1M	BPR	0.449	0.242	−46%
	NeuMF	0.422	0.178	−58%
	LightGCN	0.392	0.117	−70%
Amazon Books	BPR	0.464	0.245	−47%
	NeuMF	0.450	0.201	−55%
	LightGCN	0.382	0.111	−71%

The ceiling declines monotonically with α for all three architectures, indicating that the structural limit is universally downward in direction. However, the magnitude of this decline is strongly architecture-dependent and remains highly consistent across datasets. For example, BPR loses approximately 46–47% of its Smooth xQuAD ceiling, while NeuMF loses 55 to 58%, and LightGCN loses 70 to 71%.

This ordering aligns with the pool-depletion profiles shown in Figure 4.12. In particular, in case of LightGCN's front-depleted pool where long-tail items are already concentrated in deep positions. At low bias LightGCN has the smallest accessible supply in Z-2. Therefore, LightGCN undergoes the most severe collapse as bias increases. More generally, the base architecture emerges as a primary determinant of the attainable fairness ceiling, largely independent of the specific reranking method applied on top.

Trade-off Frontiers for NeuMF and LightGCN

Figures 4.13 and 4.14 reproduce the accuracy–fairness frontier of Section 4.4 for NeuMF and LightGCN, using the same visual encoding as for BPR.

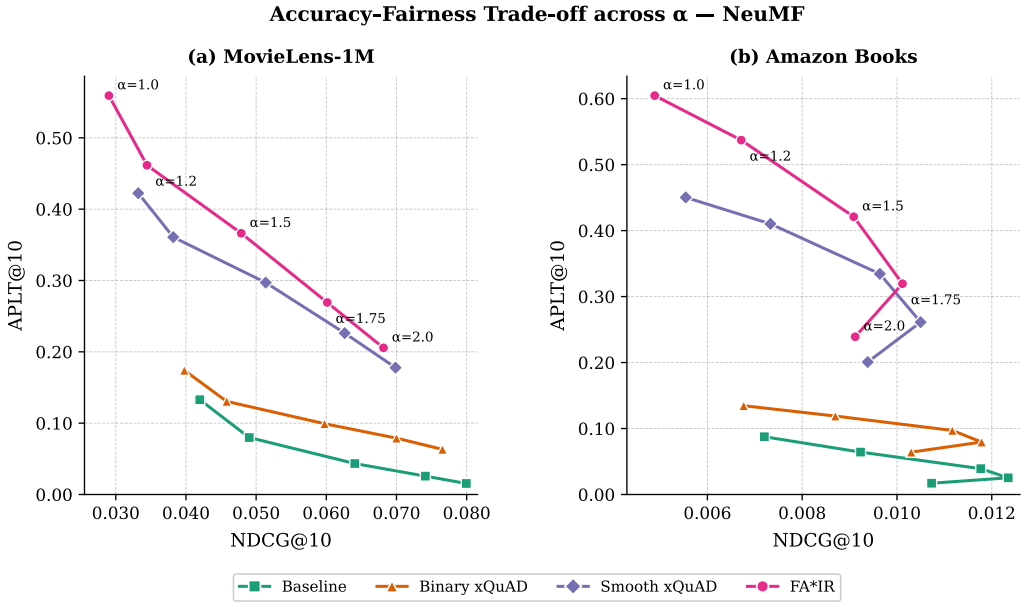


Figure 4.13: Accuracy–fairness frontier for NeuMF (same encoding as Figure 4.7). The declining staircase from $\alpha = 1.0$ (top-left) to $\alpha = 2.0$ (bottom-right) mirrors BPR.

The qualitative structure identified in Section 4.4 is preserved across both architectures: five frontier curves descending from top-left to bottom-right, with the inter-method dispersion narrowing as α increases. The resulting staircase pattern is therefore invariant in shape across architectures. What differs is its scale: LightGCN exhibits the deepest descent and strongest compression, reflecting its approximately -70% reduction in ceiling. This makes the trade-off space itself a scaled version of a common geometric structure, with LightGCN occupying the most compressed regime.

Absolute Gains and the LightGCN Reversal

The ceiling table summarizes the endpoints; Table 4.19 reports the full grid of absolute APLT gains (ΔAPLT) and relative NDCG costs (ΔNDCG%) for MovieLens-1M, and Figure 4.15 presents all conditions compactly as a heatmap. The Amazon Books grid is given in Appendix A.2 (Table A.3).

Table 4.19: MovieLens-1M: absolute APLT gain (ΔAPLT) and relative NDCG cost (ΔNDCG%) at Nash-selected parameters across all three model architectures and bias levels. At low bias LightGCN attains the largest absolute APLT gain (ΔAPLT), but this advantage reverses by $\alpha = 2.0$, where it attains the smallest. Its ΔNDCG% cost is the lowest of the three architectures at $\alpha \geq 1.5$ because its larger baseline NDCG deflates the percentage; this normalisation inflates LightGCN's apparent efficiency (Section 4.7).

α	Model	Binary xQuAD		Smooth xQuAD		FA*IR	
		ΔAPLT	ΔNDCG%	ΔAPLT	ΔNDCG%	ΔAPLT	ΔNDCG%
1.00	BPR	0.0324	5.38%	0.2583	21.73%	0.4141	33.05%
	NeuMF	0.0412	5.23%	0.2895	20.90%	0.4263	30.86%
	LightGCN	0.0560	6.81%	0.3177	20.94%	0.4322	29.32%
1.20	BPR	0.0414	6.38%	0.2774	22.98%	0.4063	33.25%
	NeuMF	0.0507	6.55%	0.2809	22.08%	0.3818	29.73%
	LightGCN	0.0623	7.05%	0.2911	18.80%	0.3735	24.50%
1.50	BPR	0.0492	6.67%	0.2627	20.57%	0.3556	27.69%
	NeuMF	0.0558	6.72%	0.2536	19.82%	0.3228	25.26%
	LightGCN	0.0615	6.42%	0.2177	14.78%	0.2612	18.00%
1.75	BPR	0.0504	6.44%	0.2310	18.81%	0.2993	23.87%
	NeuMF	0.0532	5.54%	0.2005	15.51%	0.2435	18.87%
	LightGCN	0.0527	5.27%	0.1574	10.90%	0.1827	12.72%
2.00	BPR	0.0490	5.57%	0.1972	15.77%	0.2455	19.59%
	NeuMF	0.0478	4.25%	0.1624	12.61%	0.1902	14.76%
	LightGCN	0.0423	3.78%	0.1117	7.83%	0.1266	9.10%

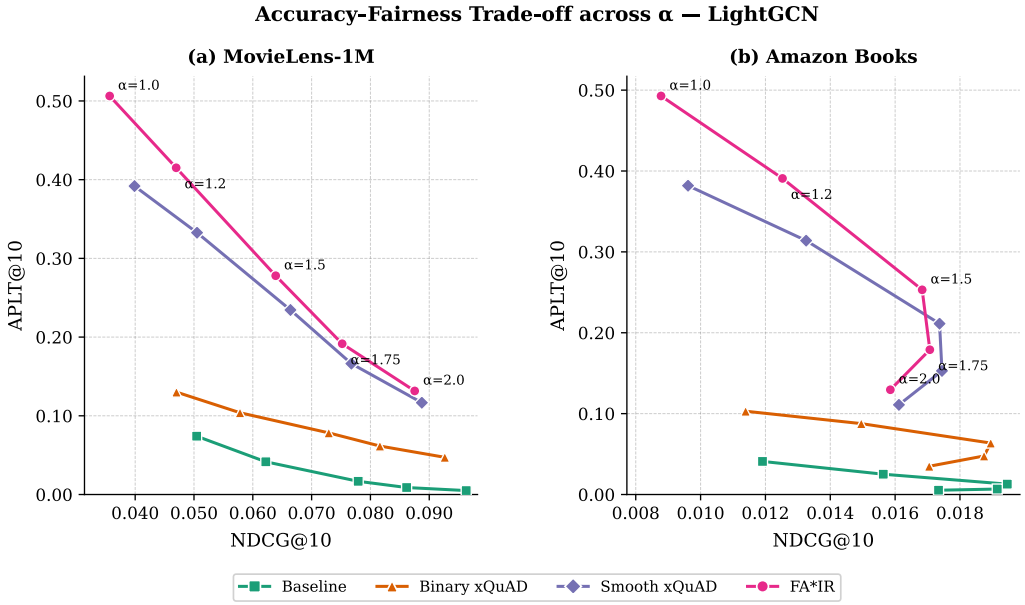


Figure 4.14: Accuracy–fairness frontier for LightGCN. The decline is the steepest of the three architectures: the $\alpha = 2.0$ frontier sits near zero APLT and the feasible region is the most compressed.

The grid reveals a reversal that is not visible in the endpoint table. At $\alpha = 1.0$, LightGCN achieves the largest absolute Smooth xQuAD gain among the three architectures (ΔAPLT $= 0.318$, compared to BPR's 0.258): when the candidate pool is still rich, LightGCN's lower initial APLT leaves more room for improvement through reranking. By $\alpha = 2.0$, this ordering inverts, with LightGCN producing the smallest gain (0.112, compared to BPR's 0.197). The same reversal is observed for FA*IR. Across the bias range, LightGCN's Smooth xQuAD gain decreases by approximately 65%, whereas BPR declines by only 24%. Thus, the architecture that benefits most from mitigation at low bias becomes the least responsive at high bias, consistent with the front-depleted structure of its candidate pool (Figure 4.12), which exhausts reachable long-tail content most rapidly.

Resolving the efficiency normalization caveat. This grid also clarifies the caveat raised in Section 4.6. Although LightGCN appeared to be the most efficient architecture, this result is sensitive to the normalization of NDCG cost. The ΔNDCG% columns show that at $\alpha \geq 1.5$, LightGCN exhibits the lowest relative NDCG loss among the three architectures (e.g., at $\alpha = 2.0$, Smooth xQuAD: LightGCN 7.8% vs. BPR 15.8%), not because

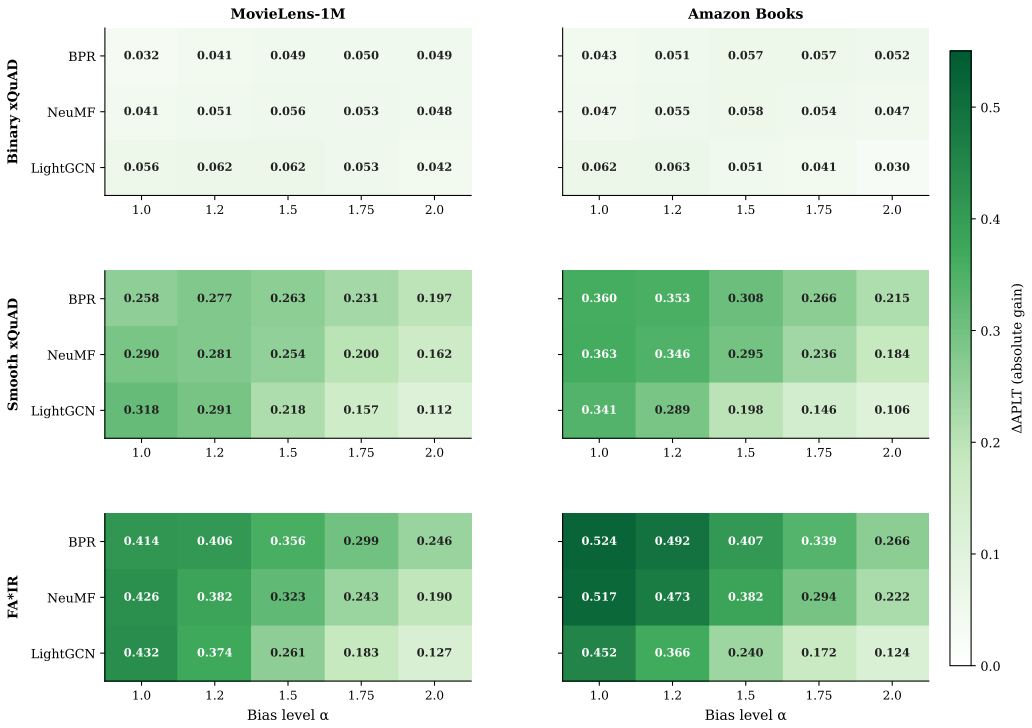


Figure 4.15: Absolute APLT gain (ΔAPLT) at Nash-selected parameters for each method (rows), architecture, and bias level, on MovieLens-1M (left) and Amazon Books (right). Darker cells denote larger gains.

it sacrifices less absolute accuracy, but because its higher baseline NDCG compresses percentage-based losses. As a result, efficiency that is defined as APLT gain divided by percentage NDCG loss is mechanically inflated for LightGCN. When evaluated in absolute terms via ΔAPLT, LightGCN is not the strongest but the weakest performer under high bias. The apparent efficiency advantage is therefore an artifact of percentage normalization, mirroring the relative-improvement distortion discussed in Section 4.6.

A Multi-Objective View

Figure 4.16 summarizes each architecture from the multi-objective perspective of Section 4.3, plotting the best Nash score (the maximum over the three methods) against α.

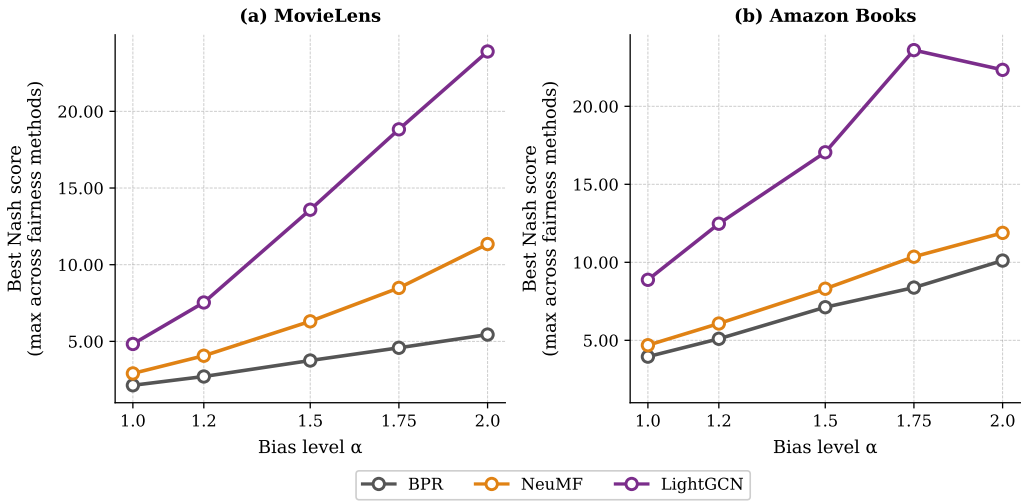


Figure 4.16: Best Nash score (maximum across the three mitigation methods) versus bias level for each architecture, on (a) MovieLens-1M and (b) Amazon Books. LightGCN's curve rises most steeply, a consequence of its collapsing baseline rather than a superior trade-off; see text.

LightGCN's best Nash score increases most sharply with α, remaining higher than both NeuMF and BPR across all bias levels. However, as established in Section 4.3, this trend should not be interpreted as LightGCN achieving a superior fairness accuracy trade-off under increasing bias. The absolute ceiling results indicate the opposite ordering. The Nash score is multiplicative in baseline-relative ratios, and because LightGCN experiences the strongest collapse in baseline APLT, it exhibits the greatest inflation in relative headroom among the three architectures. Consequently, the steeply

rising Nash curve and the steeply declining absolute ceiling represent two different projections of the same structural phenomenon.

This observation consolidates the methodological thread developed across Sections 4.3, 4.6, and 4.7: under popularity bias, all baseline-relative metrics, including the Nash score, relative improvement measures, and percentage-normalized efficiency systematically overstate performance in regimes where the absolute outcome is deteriorating most severely. In contrast, absolute APLT provides a consistent and faithful representation of the underlying structural ceiling.

Parameter Transferability

Finally, the Nash-selected parameters (Binary xQuAD $\lambda^* = 0.4$ on MovieLens-1M / 0.3 on Amazon Books, Smooth xQuAD $\lambda^* = 0.9$, FA*IR $\tau^* = 0.9$), calibrated on BPR in Section 4.3, were applied unchanged to both NeuMF and LightGCN. The ranking of methods by absolute APLT we observe that, FA*IR > Smooth xQuAD > Binary xQuAD and this is preserved across all ten architecture and dataset configurations. No architecture-specific re-tuning is required to reproduce the results.

This indicates that the Nash bargaining solution captures a property of the dataset which is bias interaction rather than being specific to the BPR base model. A single calibration therefore transfers across architectures, and together with the per-level stability observed in Section 4.3, this yields a configure-once approach that remains robust across both varying bias levels and different base models.

Chapter 5

Discussion

This chapter interprets the experimental results presented in Chapter 4 with respect to the three research questions and the hypotheses introduced in Chapter 1. Section 5.1 addresses RQ1 (the effect of bias level on mitigation effectiveness), Section 5.2 examines RQ2 (accuracy–fairness trade-offs), and Section 5.3 evaluates RQ3 (cross-architecture robustness). Section 5.4 re-assesses each hypothesis in light of the empirical findings, and Section 5.5 concludes with a consolidated summary of the discussion.

5.1 RQ1: Effect of Bias Level on Mitigation Effectiveness

All three post-processing methods, Binary xQuAD, Smooth xQuAD, and FA*IR, improve long-tail coverage (APLT@10) over the unmitigated baseline at every tested bias level and on both datasets. This confirms that post-processing reranking remains an effective fairness intervention even under strongly biased base models. However, the maximum achievable APLT@10, the *structural ceiling*, declines monotonically with α across all methods, datasets, and architectures, directly supporting Hypothesis H1.

The mechanism is the declining candidate-pool long-tail fraction (Table 4.3). At $\alpha = 1.0$, each user's top-100 candidate pool contains on average 17.8 long-tail items on MovieLens-1M; FA*IR can exploit this to meet its threshold of seven protected items in the top-10, achieving APLT ≈ 0.60. At $\alpha = 2.0$, only 4.8 long-tail items remain in the average pool, fewer than FA*IR's requirement, making the constraint infeasible for most users and reducing the ceiling to 0.29, a 52% drop that no reranking strategy can overcome without modifying the candidate pool itself. Amazon Books exhibits a similar pattern (-53% for FA*IR, -47% for Smooth xQuAD, and -76% for the BPR baseline).

Binary xQuAD represents a partial exception that clarifies this mechanism. Because it activates its diversity bonus only until a single long-tail item is placed, it requires minimal pool support and therefore remains feasible across all bias levels. Its absolute APLT gain over the baseline is consequently stable or slightly increasing with α (from +0.033 at $\alpha = 1.0$ to +0.049 at $\alpha = 2.0$ on MovieLens-1M), making it the most structurally robust method. However, this robustness comes at the cost of a lower absolute ceiling: Binary xQuAD consistently produces only a small and nearly fixed number of additional long-tail items per user, whereas Smooth xQuAD and FA*IR achieve higher gains under low bias but deteriorate sharply as the pool depletes.

The positional analysis in Section 4.5 explains this bound more precisely. The decline is driven not only by the reduction in the number of long-tail items, but by their redistribution within the pool. As bias increases, mitigation methods must increasingly source promoted items from deeper, lower-relevance regions. This migration is method-dependent and most pronounced for FA*IR, which shifts from primarily using the reachable Z-2 region (positions 11–50) to relying heavily on the deep Z-3 region (positions 51–100) at high bias across both datasets. The structural ceiling is therefore not only lower under bias but also compositionally worse: the remaining achievable exposure is increasingly constructed from items ranked least relevant by the base model, which explains why the accuracy cost of mitigation changes structurally with bias.

5.2 RQ2: Accuracy–Fairness Trade-off Under Varying Bias

The accuracy–fairness Pareto frontier contracts as α increases in two ways (Section 4.4): the achievable APLT ceiling decreases, limiting the maximum fairness gain; and inter-method separation compresses, reducing the practical impact of method choice precisely when bias is strongest. At $\alpha = 1.0$ on MovieLens-1M, FA*IR achieves a +0.41 APLT improvement at a −34% NDCG cost; at $\alpha = 2.0$, the same method achieves only a +0.25 APLT improvement at a reduced −19% NDCG cost. Both gain and cost decline, but fairness gains contract more rapidly, indicating an asymmetric degradation of the trade-off space. The feasible region of simultaneously achievable accuracy and fairness narrows as bias increases, confirming that the bias level at which the base model is trained has an asymmetric effect on both sides of the trade-off.

A key methodological result concerns the interpretation of relative met-

rics. Measures normalized against the baseline systematically inflate as bias increases because the baseline itself collapses (Sections 4.3 and 4.6). Binary xQuAD, for example, shows relative APLT improvement rising from 41% to over 420% on MovieLens-1M, while its absolute gain changes only from 0.033 to 0.049. Nash scores and percentage-normalized efficiencies exhibit the same effect, creating the illusion of improving effectiveness under bias despite a shrinking structural ceiling. Consequently, absolute APLT should be treated as the primary indicator of fairness performance, with relative metrics used only as supplementary context.

Nash bargaining parameter selection identifies a practically important finding: the optimal operating point is stable across all α levels. Binary xQuAD selects $\lambda^* = 0.4$ (MovieLens-1M) and $\lambda^* = 0.3$ (Amazon Books); Smooth xQuAD selects $\lambda^* = 0.9$ on both datasets; FA*IR selects $\tau^* = 0.9$ on both datasets. These parameters remain optimal from $\alpha = 1.0$ through $\alpha = 2.0$. Smooth xQuAD and FA*IR are fully stable across all bias levels, while Binary xQuAD shows only a minor per-dataset variation within the flat plateau region of its objective surface (Section 4.3). This indicates that optimal operating points are robust to bias variation, allowing a single calibration to remain valid without re-tuning—a practically useful property for deployment. Accordingly, Hypothesis H3 is not supported: the optimal trade-off parameter is determined primarily by the method's inherent characteristics and dataset structure, rather than by the bias level.

5.3 RQ3: Robustness Across Model Architectures

The structural ceiling holds across all three architectures, confirming its universal directionality, but its magnitude is strongly architecture-dependent and remains highly consistent across datasets. Smooth xQuAD ceiling reductions are approximately 46–47% for BPR, 55–58% for NeuMF, and 70–71% for LightGCN as α increases from 1.0 to 2.0 (Table 4.18). LightGCN, despite achieving the highest baseline NDCG, exhibits the steepest fairness collapse, revealing a systematic accuracy–fairness inversion at the architecture level: LightGCN achieves the best recommendation accuracy but the worst long-tail ceiling; BPR achieves the lowest accuracy but the highest long-tail ceiling; NeuMF lies between the two on both dimensions. This inversion is consistent across both datasets and all five α levels, indicating it is a structural property of how each architecture learns representations rather than an artifact of experimental conditions.

The mechanism is explained by the pool-depletion profiles in Section 4.7. LightGCN's graph propagation amplifies popular-item representations through multi-hop neighbourhood aggregation, producing a candidate pool that is strongly front-depleted: long-tail items are concentrated in deep positions even at low bias, which leads to rapid exhaustion of the reachable supply as α increases. This directly explains the ceiling ordering: BPR's more uniform pool distribution provides the greatest headroom for reranking, while LightGCN's front-depleted pool exhausts reachable long-tail candidates earliest.

This also explains a reversal not visible in endpoint summaries (Section 4.7). At $\alpha = 1.0$, LightGCN yields the largest absolute Smooth xQuAD gain among the three architectures (ΔAPLT $= 0.318$, vs. BPR's 0.258): when the pool is still rich, LightGCN's lower initial APLT leaves more room for improvement. By $\alpha = 2.0$, this ordering inverts, with LightGCN producing the smallest gain (0.112 vs. 0.197), reflecting a 65% collapse compared to BPR's 24%. The architecture that benefits most from mitigation at low bias thus becomes the least responsive at high bias.

Efficiency-based interpretations must be treated with care. LightGCN appears most efficient when measured as APLT gain per percentage NDCG loss, but this is driven by its higher baseline NDCG compressing percentage denominators rather than genuine trade-off superiority. In absolute terms, it is the weakest performer under high bias, consistent with the relative-metric distortions discussed in RQ2.

Finally, Nash-selected parameters transfer unchanged across architectures, and the ranking FA*IR > Smooth xQuAD > Binary xQuAD by absolute APLT is preserved in all ten architecture–dataset combinations. The Nash bargaining solution therefore reflects a dataset–bias interaction rather than a BPR-specific calibration, yielding a single configuration that generalizes across both varying bias levels and different base models (Section 4.7).

5.4 Revisiting the Hypotheses

H1: *The effectiveness of fairness-aware reranking methods decreases as popularity bias α increases.* This is **confirmed**. All methods exhibit a declining APLT ceiling as α increases, driven by the pool long-tail fraction declining from approximately 17% to 5%. No post-processing method can overcome this supply-side constraint without modifying the candidate pool itself.

H2: *Different reranking strategies exhibit varying robustness to increasing bias levels.* This is **confirmed with nuance**. Binary xQuAD is the most structurally robust due to its minimal pool requirement. FA*IR exhibits the

largest absolute ceiling decline but retains a consistent advantage over Smooth xQuAD across all bias levels. Critically, architectural choice is a stronger determinant of structural robustness than the choice of post-processing method: the base model's candidate pool distribution fundamentally constrains the ceiling within which all methods operate.

H3: *The optimal fairness–utility trade-off parameter is dependent on the underlying bias distribution.* This is **not supported**. Nash-optimal parameters remain stable across all α levels on both datasets and transfer across architectures, indicating that the optimal trade-off point is governed primarily by method-intrinsic characteristics and dataset structure rather than by the level of popularity bias.

5.5 Summary

- Post-processing methods improve long-tail coverage at all bias levels, but the achievable ceiling declines monotonically with α. This is driven by the shrinking long-tail fraction in the candidate pool and the resulting forced migration of promoted items into deeper, lower-relevance regions.

- The decline is architecture-dependent in magnitude but universal in direction: all models and methods are constrained by the same upstream structural limit. Architectures with front-depleted pools (notably LightGCN) experience the steepest collapse and can reverse from being the most responsive to the least responsive to mitigation as bias increases.

- Baseline-relative metrics, including relative improvement, Nash scores, and percentage-normalized efficiency, systematically inflate under bias because the baseline itself collapses. Absolute APLT is the only metric that faithfully reflects the structural ceiling and should be treated as the primary measure in this setting.

- Nash-selected parameters remain stable across bias levels and transfer across architectures, providing a single-calibration solution that is robust in practice.

- Since the limiting factor lies in the candidate pool upstream of any reranking stage, mitigating popularity bias at the source, through data curation, training-time objectives, or improved data collection, is necessary to restore sufficient long-tail availability for post-processing methods to remain effective.

Chapter 6

Conclusion

6.1 Overview

This thesis investigates a question that prior work on fairness-aware recommender systems has largely left unresolved: how the effectiveness of post-processing mitigation methods changes when the underlying base model is trained under varying levels of popularity bias. While existing literature has established that post-processing reranking can improve long-tail exposure, it has typically evaluated this under fixed dataset conditions, without analyzing how degradation in the base model constrains the maximum achievable fairness improvement. We address this gap through a controlled two-stage experimental framework in which popularity bias is systematically induced at five levels ($\alpha \in \{1.0, 1.2, 1.5, 1.75, 2.0\}$) using power-law interaction resampling, applied to two benchmark datasets (MovieLens-1M and Amazon Books) and evaluated across three recommendation architectures (BPR, NeuMF, LightGCN).

The central contribution of this thesis is the identification and characterization of the *structural ceiling*: a hard, data-driven upper bound on long-tail promotion that declines monotonically with the level of base-model bias, independent of the post-processing method applied. This ceiling arises not from limitations in the reranking algorithms themselves, but from the progressive depletion of long-tail items in the candidate pool as the base model increasingly concentrates its representations on head items.

6.2 Main Findings

Post-processing reranking remains a viable fairness intervention even under substantial popularity bias: all three methods improve long-tail coverage over

the unmitigated baseline at every α level and on both datasets. However, the magnitude of this improvement is bounded by a structural ceiling that shrinks monotonically with α, driven by the progressive depletion of long-tail items in the candidate pool. The ceiling is universal in direction but architecture-dependent in severity: more expressive base models produce more front-depleted candidate pools and therefore experience steeper ceiling collapses. The choice of post-processing method does not alter the direction of this decline, only its magnitude.

A secondary finding with practical relevance is the stability of Nash-selected parameters. The optimal trade-off settings (Smooth xQuAD $\lambda^* = 0.9$, FA*IR $\tau^* = 0.9$, Binary xQuAD $\lambda^* \in \{0.3, 0.4\}$) remain unchanged across all five bias levels and transfer without modification across all three architectures. A single calibration is therefore sufficient for deployment across varying bias conditions and base models.

6.3 Limitations

Several limitations bound the scope of the conclusions. First, the bias induction procedure produces a controlled but artificial gradient; real-world popularity bias emerges from complex feedback dynamics between users, items, and recommendation systems that power-law resampling cannot fully replicate. The extent to which the structural ceiling finding holds in environments where bias is driven by genuine user behavior remains an open question.

Second, the experiments rely on two datasets from the domains of film and book recommendation. Other domains, such as news, music, or employment recommendation, may exhibit different long-tail distributions and user-item interaction patterns that affect both the severity of pool depletion and the effectiveness of reranking methods.

Third, evaluation is conducted at a single point in time. In production systems, popularity bias evolves continuously through exposure feedback loops. Whether Nash-selected parameters remain stable as the bias distribution drifts over time is not addressed in this work.

Finally, the evaluation framework measures APLT@10 using a fixed long-tail definition derived from the $\alpha = 1.0$ distribution. This definition is consistent and enables meaningful cross-α comparison, but alternative definitions, such as item-level entropy or relative exposure measures, may capture different aspects of fairness and could yield different structural patterns.

6.4 Closing Remarks

This thesis set out to quantify the limits of post-processing popularity bias mitigation under controlled conditions. The answer is unambiguous: post-processing methods can improve long-tail coverage meaningfully, but they operate within a structural ceiling that shrinks as the base model becomes more biased. This ceiling is not a property of any particular reranking algorithm; it is a property of the candidate pool, and no algorithm can overcome it without access to additional long-tail items at the retrieval stage.

The practical implication is equally clear. Fairness-aware reranking is a valuable and deployable tool, but it is not a substitute for addressing popularity bias at its source. A system that relies solely on post-processing to compensate for a highly biased retrieval model will find that the fairness headroom available to it is exactly the headroom left by the retrieval model itself, and that headroom shrinks with every increase in bias. Effective long-tail promotion therefore requires a coordinated approach: bias mitigation at the training stage to preserve tail items in the candidate pool, and targeted post-processing at the reranking stage to fully exploit the preserved pool.

Long-tail items cannot be promoted if they cannot first be found.

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Appendix A

Supplementary Results

This appendix collects the per-architecture and per-dataset result tables that are summarized in the main text. They are provided for completeness and reproducibility; the conclusions drawn in the experiments and results chapter and its discussion rely only on the aggregated tables presented inline there.

A.1 Per-Architecture Mitigation Efficiency

Section 4.6 reports mitigation efficiency averaged across the three base architectures (Table 4.15). Tables A.1 and A.2 give the full per-architecture breakdown for MovieLens-1M and Amazon Books respectively. As discussed in Section 4.7, LightGCN's higher nominal efficiency at most bias levels reflects its larger baseline NDCG deflating the percentage-normalized denominator rather than a genuinely more favorable trade-off; the absolute APLT gains in Table 4.19 tell the opposite story.

A.2 Cross-Architecture Gains: Amazon Books

Section 4.7 presents the absolute APLT gains and relative NDCG costs across architectures for MovieLens-1M (Table 4.19) and shows the full grid for both datasets compactly in Figure 4.15. Table A.3 provides the corresponding numerical grid for Amazon Books. The architecture reversal described in the main text — LightGCN yielding the largest absolute gains at low bias and the smallest at high bias — is reproduced here.

Table A.1: MovieLens: mitigation efficiency (APLT gain per 1% NDCG cost) at Nash-selected parameters. Higher values indicate a more favourable accuracy–fairness trade-off; the highest value in each row is bolded. Within each method, LightGCN attains the highest nominal efficiency at most bias levels, but this reflects its larger baseline NDCG inflating the percentage-normalised denominator rather than a genuinely better trade-off (see Section 4.7).

α	Method	BPR	NeuMF	LightGCN
1.00	Binary xQuAD	0.0060	0.0079	**0.0082**
	Smooth xQuAD	0.0119	0.0139	**0.0152**
	FA*IR	0.0125	0.0138	**0.0147**
1.20	Binary xQuAD	0.0065	0.0077	**0.0088**
	Smooth xQuAD	0.0121	0.0127	**0.0155**
	FA*IR	0.0122	0.0128	**0.0152**
1.50	Binary xQuAD	0.0074	0.0083	**0.0096**
	Smooth xQuAD	0.0128	0.0128	**0.0147**
	FA*IR	0.0128	0.0128	**0.0145**
1.75	Binary xQuAD	0.0078	0.0096	**0.0100**
	Smooth xQuAD	0.0123	0.0129	**0.0144**
	FA*IR	0.0125	0.0129	**0.0144**
2.00	Binary xQuAD	0.0088	**0.0112**	0.0112
	Smooth xQuAD	0.0125	0.0129	**0.0143**
	FA*IR	0.0125	0.0129	**0.0139**

Table A.2: Amazon Books: mitigation efficiency (APLT gain per 1% NDCG cost) at Nash-selected parameters. Higher values indicate a more favourable accuracy–fairness trade-off; the highest value in each row is bolded. As on MovieLens-1M, LightGCN's higher nominal efficiency reflects its larger baseline NDCG inflating the percentage-normalised denominator rather than a genuinely better trade-off (see Section 4.7).

α	Method	BPR	NeuMF	LightGCN
1.00	Binary xQuAD	0.0064	0.0077	**0.0142**
	Smooth xQuAD	0.0142	0.0157	**0.0177**
	FA*IR	0.0147	0.0161	**0.0172**
1.20	Binary xQuAD	0.0074	0.0095	**0.0144**
	Smooth xQuAD	0.0153	0.0168	**0.0190**
	FA*IR	0.0154	0.0173	**0.0184**
1.50	Binary xQuAD	0.0100	0.0114	**0.0193**
	Smooth xQuAD	0.0161	0.0163	**0.0185**
	FA*IR	0.0164	0.0168	**0.0178**
1.75	Binary xQuAD	0.0107	0.0119	**0.0192**
	Smooth xQuAD	0.0148	0.0158	**0.0164**
	FA*IR	0.0151	**0.0163**	0.0159
2.00	Binary xQuAD	0.0109	0.0118	**0.0179**
	Smooth xQuAD	0.0144	0.0146	**0.0150**
	FA*IR	0.0147	**0.0148**	0.0145

Table A.3: Amazon Books: absolute APLT gain (ΔAPLT) and relative NDCG cost (ΔNDCG%) at Nash-selected parameters across all three model architectures and bias levels. As on MovieLens-1M, LightGCN attains the largest absolute APLT gain at low bias but the smallest by $\alpha = 2.0$, and its lower ΔNDCG% at high bias reflects percentage normalisation against a larger baseline NDCG rather than a genuinely cheaper trade-off (Section 4.7).

α	Model	Binary xQuAD		Smooth xQuAD		FA*IR	
		ΔAPLT	ΔNDCG%	ΔAPLT	ΔNDCG%	ΔAPLT	ΔNDCG%
1.00	BPR	0.0425	6.62%	0.3603	25.39%	0.5240	35.66%
	NeuMF	0.0470	6.07%	0.3627	23.03%	0.5171	32.13%
	LightGCN	0.0621	4.37%	0.3410	19.28%	0.4519	26.27%
1.20	BPR	0.0512	6.88%	0.3535	23.17%	0.4924	31.94%
	NeuMF	0.0546	5.76%	0.3459	20.59%	0.4729	27.28%
	LightGCN	0.0626	4.34%	0.2888	15.22%	0.3657	19.88%
1.50	BPR	0.0570	5.71%	0.3084	19.20%	0.4071	24.83%
	NeuMF	0.0579	5.09%	0.2953	18.09%	0.3819	22.78%
	LightGCN	0.0508	2.63%	0.1984	10.74%	0.2402	13.49%
1.75	BPR	0.0566	5.29%	0.2665	17.99%	0.3394	22.44%
	NeuMF	0.0542	4.55%	0.2360	14.97%	0.2942	18.07%
	LightGCN	0.0409	2.13%	0.1461	8.91%	0.1723	10.86%
2.00	BPR	0.0520	4.78%	0.2154	14.95%	0.2660	18.10%
	NeuMF	0.0471	4.01%	0.1837	12.56%	0.2220	15.03%
	LightGCN	0.0296	1.66%	0.1056	7.05%	0.1242	8.57%

Table A.4: Bias statistics for the supplementary debiasing experiment. The natural distribution ($\alpha = 1.0$) is included for reference.

α	Gini coefficient	Candidate-pool LT (%)	Baseline APLT@10
0.50	0.320	30.7	0.340
0.75	0.453	23.7	0.254
1.00	0.562	17.8	0.191

A.3 Behaviour Under Bias Reduction

To complement the main experiments, we evaluate two debiasing settings ($\alpha = 0.5$ and $\alpha = 0.75$), which flatten the popularity distribution relative to the natural dataset ($\alpha = 1.0$).

This analysis is limited to MovieLens-1M with BPR and is intended as a sanity check rather than a full replication of the experimental pipeline.

Table A.4 and Figure A.1 show that reducing α decreases popularity concentration and increases the proportion of long-tail items in the candidate pool. The Gini coefficient drops from 0.562 at $\alpha = 1.0$ to 0.453 and 0.320, while candidate-pool long-tail coverage increases from 17.8

Correspondingly, baseline BPR recommendations become more long-tail oriented, with APLT@10 increasing from 0.191 to 0.254 and 0.340.

Figure A.2 extends the structural-ceiling analysis to $\alpha < 1$. The results follow the same general trends observed in the main experiments, with mitigation effectiveness remaining dependent on the composition of the candidate pool.

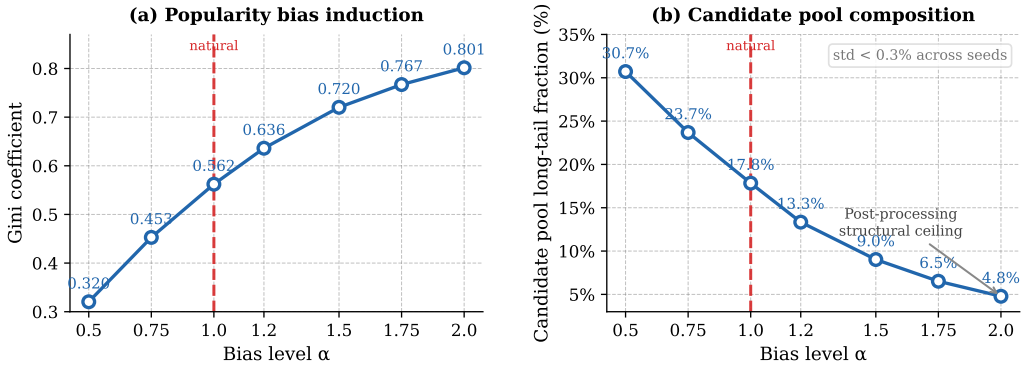


Figure A.1: Popularity-bias induction statistics extended to include debiasing settings ($\alpha < 1$). The monotonic relationship between popularity concentration and candidate-pool long-tail availability persists across the full $\alpha = 0.5$–2.0 range.

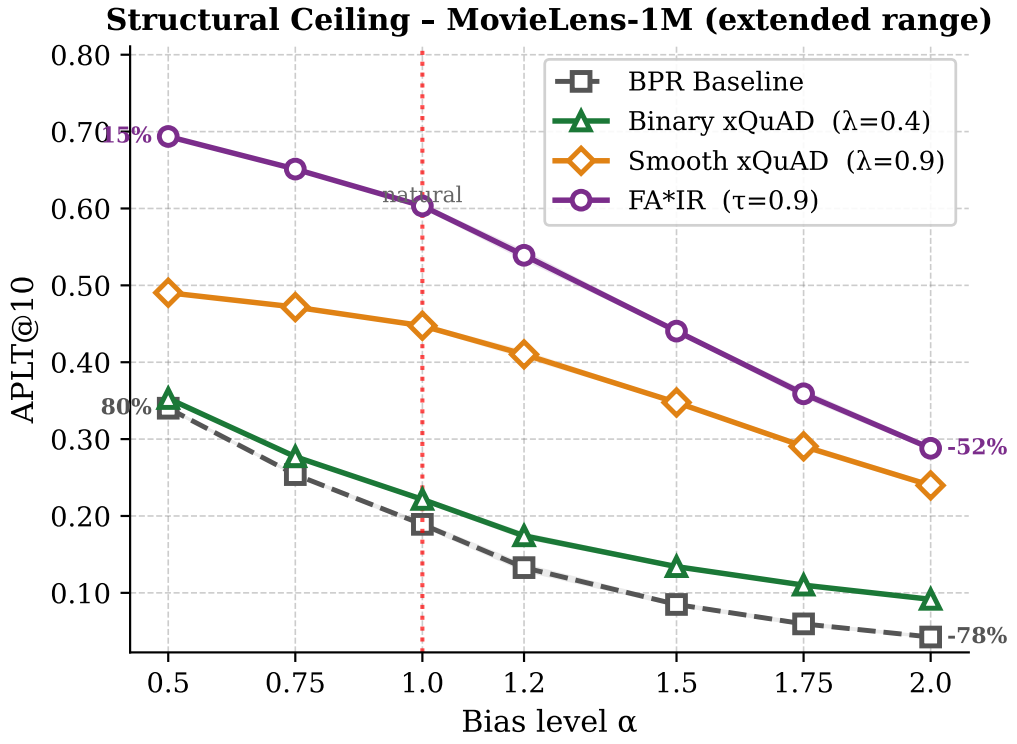


Figure A.2: Structural-ceiling analysis across the full $\alpha = 0.5$–2.0 range for BPR on MovieLens-1M. Mitigation headroom narrows toward both extremes of the bias spectrum, but for different reasons: candidate scarcity under high bias and reduced improvement potential under strong debiasing.